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Abstracts

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together with

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Morphological reanalysis and the Old English *minor* paradigm

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The paper addresses the issue of morphological constitution of the early English nominal inflection, focusing on the developments which contributed to its later restructuring. One of the prominent features of the early English inflection was an evident tendency, revealed by nouns considered minor (unproductive) to adopt the inflectional endings of the productive types. This marked inclination of some nouns can be particularly well seen in the formations containing the original *u*-stems, which constituted a separate group in Germanic, known in the standard grammars as *u*-declension. The available textual material proves that members of this relatively small group of nouns tend to fluctuate between the inherited and innovative paradigmatic patterns, testifying thus to a growing instability of this declensional type. The fluctuation between the two competing types can be seen in forms of the nominative/accusative plural of masculine paradigm, where alongside the expected *-u* ending forms in *-as*, extended from the productive *a*-paradigm, are attested (*sunu* ~ *sunas*, *wudu* ~ *wudas*). Similarly, the genitive and dative singular endings *-a* and *-e* (respectively) regularly alternate with the innovative *-es* and *-e* markers (*felda* ~ *feldes* (gen.), *felda* ~ *felde* (dat.)). Traces of this progressing tendency can also be detected in feminine stems, where the competition takes place between the inherited inflection and the expanding strong feminine inflection. Such lack of stability in the Old English inflectional paradigms attests to an ongoing syncretism, resulting in the eventual disintegration of the original stem type distinctions.

The analysis, conducted on the *Dictionary of Old English Electronic Corpus*, is intended to be both a qualitative and quantitative study of the Old English *u*-stems. Aimed at presenting a systematic account of the newly emerging tendency, the investigation will seek to determine the extent and pattern of dissemination of the productive inflectional endings in the original *u*-stems, paying due attention to the niceties of the process of gradual reorganisation of the minor paradigm, its consequences in the later inflectional system of English as well as its theoretical implications.

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Homophonous morphemes in nominal and verbal morphology: Implications for the reconstruction of the Pre-Basque type

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Modern Basque (and historical Basque as a whole; oldest texts start in the 16th century) is a XXIV-type language in Greenberg's (1966) classification; that is, a non-rigid SOV language with Gen-N order and postpositions, but with N-Adj order. The nominal morphology of Basque, including its case-marking, is overwhelmingly suffixing. However, Basque's verbal morphology includes both suffixes and prefixes. In addition, Basque has a very rich morphology, whereby ABS, ERG, and DAT arguments are double-marked in nominal and verbal morphology. Within verbal agreement, ABS person-markers appear as prefixes, while ERG and DAT markers are generally suffixes.

In order to explain the presence of suffixes and prefixes in the verb, a group of scholars (Trask 1977, Gómez & Sainz 1995, Lakarra 2006) have assumed that, at some ancient stage, Pre-Basque used to be a head-initial language with VO order. According to this hypothesis, at this early stage a number of prefixes got attached to the verb. Then, at a later stage, Pre-Basque turned towards an OV order —perhaps by contact with Latin—, and at this stage it started to take suffixes in both the verb and the NP.

While the hypothesis that at an early stage Pre-Basque may have been a VO language is very difficult to falsify, I would like to explore an alternative scenario. What is crucial in this alternative approach is to notice that some of the morphemes in the verbal morphology of Basque have homophonous morphemes in the nominal morphology. Most scholars agree that these morpheme-doublets may have a common origin. If this is correct, i.e. if these doublets are related in origin, it is more plausible for the verbal morphemes to have originated from the corresponding nominal morphemes than the other way around. According to Nichols (1992), morpheme migration from dependent-marking to head-marking is commonly attested, but the opposite direction in morpheme migration is practically unattested.

Based on this evidence, we may have an early stage of Pre-Basque with only or mostly nominal morphology. This was a dependent-marking language. Then, at a later stage —perhaps due to contact or because of peripherality (Nichols 1992)—, Pre-Basque took strong head-marking tendencies regarding the verb. Thus, a fair number of nominal suffixes migrated to the verb, mostly as prefixes (or infixes, in 2nd position) but also as suffixes. At this head-marking stage, Pre-Basque also created a very rich agreement pattern on the verb, giving rise to a double-marking pattern for the clause. This person agreement pattern was evenly distributed between prefixes and suffixes.

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Linguistic self-regulation: the case of Greek gender change in progress

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The literature on linguistic change has considered for the last decades the inter-disciplinarity between historical linguistics, sociolinguistics, psycholinguistics, etc., having as its basic aim to study inherent variation in language use and, as a result, to prove linguistic change in progress. This paper reports on the results of a statistical analysis of natural discourse data, which reveal a morphological change in progress.

Specifically, in the Greek language one can find nouns that tend to change their gender due to morphological pressure or due to obedience to the principles of prototypicality and neutralisation (Anastassiadis-Symeonidis & Chila-Markopoulou: 2003). Nouns that end in –os are prototypically masculine, e.g. ο φίλος [o 'filos] 'the friend', ο δρόμος [o 'dromos] 'the street'. Nevertheless, some Greek words with the same suffix are feminine or neutral (as the article before the noun reveals) e.g. η ψήφος [i 'psifos] 'the vote', η άμμος [i 'amos] 'the sand' – το κράτος [to 'kratos] 'the state', το μέλος [to 'melos] 'the member'. However, abiding by the prototypical rule of gender, the feminine nouns tend to be used as masculine ones, especially in accusative plural where the phonetic environment is suitable, e.g. τους φίλους [tus 'filus] 'the friends', τους δρόμους [tus 'dromus] 'the streets'. Such a variant is being widely used, even by some representatives of the media.

Based on the previous theoretical framework on gender assignment, we will present: a) evidence for the existence of a morphological change in progress, b) the motivation of this change and c) some basic hypotheses about the way this change is being diffused, bearing in mind that linguistic change is gradually spread.

Our data is derived from electronic texts (via the Internet) and from recorded news broadcasts. The analysis of the instances of the nouns that undergo gender change gave us the following results: a) the phonetic similarity of the masculine article and inflectional morpheme motivates change, b) the change is being consolidated by high frequency of the word, especially in its accusative form which is both more frequent and phonetically/morphologically similar, g) similarity and frequency influence the subjects' judgements of acceptability (see also Bybee: 2006), allowing a wide use of the nouns with new genders, d) due to its high frequency the noun is resistant to change in formal conversational contexts (e.g. state media), e) young people tend to use the less prestigious new gender form, and f) left-wing parties, considered to be progressive in their politics, exhibit a wide use of the nouns in their new gender form.

In conclusion, after analysing the reason and the process that leads to this kind of morphological change in progress, we outline the theoretical framework of self-regulation, according to which use is imposed on linguistic structure, restructuring the linguistic system.

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Lexical derivation in Mandarin Chinese

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In the Chinese language, morphologically complex words have been attested since the remote diachrony of the language, including both stem-modifying processes and agglutination of morphemes, mostly lexical and free in the classical language (see Baxter & Sagart 1998); the question of the distinction between (lexicalized) phrases, compounds and derived words is still under debate today (see e.g. Pan W., Ye B. & Han Y. 2004). In our talk we shall focus on the application of the notions of “derivation” and “affix” to Chinese, showing that it is possible to analyse as instances of grammaticalized (proto-)derivational formants the right-hand elements in word formation schemas (Booij 2005) such as [[X]N [学 *xué*]]N “branch of science concerned with X” (心理学 *xīnlǐxué* “psychology”) or [[x]X [性 *xìng*]]N “the quality of X / connected with X” (抽象性 *chōuxiàngxìng* “abstractness”).

In these cases we cannot rely mainly on formal criteria, being there no difference in the phonological and orthographical shapes of the “affix” and the equivalent lexeme (see Bisang 1996); however, for such constituents we have diachronical evidence of meaning shifts either reducing the number of distinctive features conveyed by the morpheme or isolating one of its meanings, ideally a peripheral one (*generalizing* and *isolating abstraction*; Heine, Claudi & Hünemeyer 1991) which may be regarded as a “diagnostic test” of grammaticalization. Such patterns of semantic evolution are well attested in SAE languages (e.g., O.H.G. *heit* “person, sex, condition, rank, quality” > O. Eng. *hād* > Mod. Eng. *-hood*) and may sometimes be visible even in the synchronic stage (Du. *boer* “farmer” > *-boer* “trader in X”; *sigaren-boer* “cigar seller”, Booij 2005); these facts seem to support our analysis, namely that the grammaticalization of a lexeme (typically a compound constituent) into a (proto-)derivational marker may be observed not only in morphologically rich languages such as the inflectional ones, but also in isolating languages like Chinese, if we take into consideration historical semantics, thus making it possible to define the notions of “affix” and “(lexical) derivation” in a cross-linguistically consistent way.

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Sign language morphology, typology, and universal grammar, or how to ride a bicycle

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Wray and Grace (2007) begin their critique of nativist theories of universal grammar with an observation: “until the early 19th century, no one would have known – or cared – that humans possess the ‘universal’ capacity to ride a bicycle after a little practice.” (p. 544) The lesson to take from their observation is that the bicycle succeeds because it takes advantage of human traits. Humans invent the cultures and cultural artifacts that we do because we have traits that help these cultures and artifacts to survive. By studying the cultures and artifacts we learn about the traits. And language, more than any other cultural artifacts, depend on a certain narrow set of human traits and thus provide a window on human nature.

Viewing languages as cultural artifacts that depend on human nature leads to a novel nativist view of language: none of the traits that languages depend on need be dedicated to language. They are simply that set of traits that permits languages to be created. This view makes no claim that language arises from general computational ability but it does not demand a dedicated language organ or faculty either (Anderson and Lightfoot 2002, Everett 2006).

It is congenial to the study of sign languages and especially sign language morphology, which is typologically very different from spoken language morphology (Aronoff et al. 2005, 2008). The roots of this typology lie in the visuospatial realm: sign languages take advantage of space in their morphology because people have eyes and they lie in the front of their heads. Spoken languages also depend on vision, inasmuch as they all use pointing together with deictic expressions. The sentence “she’s over there” or its equivalent in any other language makes no sense unless the speaker points to a referential locus that the addressee can see. Still, spoken languages never grammaticalize a morphological spatial agreement system of referential loci in the way that sign languages do.

Sign languages also use the signer’s body, first to represent the subject of a sentence (Meir et al. 2007) and then later to represent the first person as opposed to others. Both these uses of the body are iconic but the second involves the development of a syntactic person system and the creation of a morphological representation of this system that depends on moving the subject marker off the body.

Thus, detailed study of the morphology of a variety of sign languages show how human languages make use of those human traits that are available to them. The difference in media, speech vs. sign, together with detailed analysis of individual systems within these media, permit us to understand how humans make language from what nature gives them.

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Paradigmatic structure in production and comprehension

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This paper addresses the role of paradigmatic structure in production and comprehension. In my presentation, I will first focus on speech production. In the model of speech production proposed by Levelt, Roelofs and Meyer (1999), inflectional paradigms do not play a role. Inflectional features are specified at the level of a word's lemma. Nouns, for instance, carry a feature for number, which can be either singular or plural in a language like Dutch. Given a word's lemma and its specification for number, the form of its stem and the appropriate number morpheme are activated, and the phonological spell-out of the appropriate inflected form proceeds independently and encapsulated from all other word forms in the lexicon.

The study by Levelt et al. (1999) discusses unpublished experiments addressing the production of noun singulars and plurals using the picture naming paradigm. Plural-dominant nouns, nouns for which the plural form is more common than the singular form, elicited longer picture naming latencies than singular-dominant nouns. This inhibitory effect of dominance affected plural and singular nouns to the same extent. Levelt and colleagues interpreted these results as indicating competition between singular and plural lemma representations, which would exist side by side for plural-dominant nouns such as EYE and EYES, but which would be absent for singular-dominant nouns like NOSE.

Further inspection of these data show, however, that the informational complexity of the number paradigm, as gauged by Shannon's entropy measure, is at issue. The greater the entropy, which is the same for the singular and plural form of a given noun, the longer the picture naming latencies tend to be for these inflected forms. The inhibitory effect of entropy in picture naming contrasts with the facilitatory effect of plural dominance in visual and auditory comprehension (Baayen, Dijkstra & Schreuder, 1997), and bears witness to the role of item-specific inflectional probabilities in speech production.

In my presentation, I will first review the original experiments mentioned by Levelt et al. (1999) together with a recent replication study with improved experimental materials. I will then discuss the theoretical implications of our results for the role of paradigmatic structure in speech production against the background of a series of experimental studies that inform us about the role of paradigmatic structure in language comprehension (e.g., Milin et al, submitted).

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Origin and development of defective paradigms

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Defectiveness in inflectional paradigms is a serious problem for morphological theory -- the acquisition and formal representation of lexically specified gaps are the subject of intense debate. But a full appreciation of the synchronic mechanisms must also take into account the diachronic dimension: how do gaps work their way into the system in the first place, and what further paradigmatic changes do they induce? We offer a DIACHRONIC typology of defectiveness, distinguishing the various ways it arises, which in turn will illuminate our understanding of synchrony.

The chief source of defective paradigms is the lexicalization of disturbances within individual components of grammar. **Semantic** constraints may become conventionalized even where the original motivation has become opaque, as often with pluralia tantum. **Phonological** constraints may likewise be incorporated into morphology: certain phonological configurations are banned, but only for selected lexemes. Unresolved competition between different **morphological** variants (typically stem alternants) may be codified as a gap in those paradigmatic cells subject to allomorphy. A second route to defectiveness is through inertia following word class change, e.g. denominal adjectives or verbs may remain only partially assimilated to the new class. A third route involves morphological mismatches (deponency), where the link between morphosyntactic values and morphological form is skewed, depriving some values of their proper morphological expression.

The potential for further diachronic development varies. Defectives that arose through the first two pathways ultimately yield to the default pattern. Defectives of the third type (mismatches) have no obvious recourse to the default, and any resolution of the gap must involve deformation of the paradigm through **syncretism**, **polarity** or outright **innovative affixation**.

Examples are drawn from Dagestanian, Dravidian, Indo-European, Muskogean, Oceanic Oto-Manguean, Pama-Nyungan, Semitic and Ubangian languages.

Lexicalisation of suffixes in Modern Danish: variation and change in prosodic morphology

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Basbøll (2005: 351-513) is *both* a comprehensive presentation of a general cross-linguistic model for word structure based upon systematically graded productivity of endings, or -- in other words -- different degrees of integration of suffixes with their stems, *and* a detailed application of this model to modern Danish whose prosodic morphology presents the challenge of combining the general Germanic stress and intonation with the stød (a laryngeal syllabic rhyme prosody, see Basbøll 2005: 82-87, 265-292, 318-321, Grønnum & Basbøll 2007). The talk will present further developments of the lexical and morphological parts of the theory. I shall discuss -- with particular reference to Danish but taking also the cross-linguistic model into consideration -- a possible answer to each of the following questions:

- 1) Which difference is there between a lexicalised suffix and a lexicalised word form?
- 2) Which suffixes can be lexicalised?
- 3) What is the resulting word structure if a suffix is lexicalised?
- 4) Is there a discovery procedure for identifying lexicalised suffixes?
- 5) Is zero a suffix, and if so, can it be lexicalised?

Implications of these answers for variation and change in morphology will be discussed, in particular the variability of certain inflectional suffixes and the possibility of an ongoing change (Grønnum 2005, Grønnum & Basbøll 2007: 203-205) in the Non-Stød Principle (Basbøll 2003, 2005: 379-383).

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Variation in Exocentric Compounds

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The notion of a bahuvrihi compound goes back to the ancient Sanskrit grammarians, and the broadened notion of an exocentric compound is familiar from Bloomfield's *Language*. In some languages, effects similar to those achieved by exocentric compounds are achieved by affixal morphology, e.g. Kayardild has an affix which could be glossed as 'having ~', and Lezgian has one which could be glossed as 'not having ~'. Among the languages that are described as having exocentric compounds, a surprisingly varied set of structures is found (see the table below for a sample, which omits some of the apparently more marginal types). Despite the apparently clear distinction between endocentric and exocentric compounds, the borderline is rather fuzzy, and we might argue about the exocentricity of some of the categories listed, though linguists have described all of these types as exocentric. In this paper the focus is on the variety of structures, their distribution, and the factors which might lead to the rarity of some of the structures.

Type	Categories	Example	Also found in
Bahuvrihi	$N + N \rightarrow N$	<i>Kayardild</i> face + woman $\rightarrow$ effeminate looking man	Danish, English, Fongbe, Pawnee
Bahuvrihi	$A + N \rightarrow N$	<i>Kayardild</i> water + bad $\rightarrow$ place with bad water	English, Fongbe, French, German, Hausa, Kashmiri, Pawnee
Bahuvrihi	$A + N \rightarrow A$	<i>Persian</i> little + speech $\rightarrow$ quiet	Indonesian, Kashmiri
Bahuvrihi	$A + N \rightarrow V$	<i>Thai</i> good + heart $\rightarrow$ be glad	Wolof
Bahuvrihi	$Num + N \rightarrow N$	<i>Thai</i> three + edge $\rightarrow$ triangle	German
Bahuvrihi	$V + N \rightarrow N$	<i>Urarina</i> be bald + head $\rightarrow$ bald person	English
Superordinate	$N + N \rightarrow N$	<i>Vietnamese</i> table + chair $\rightarrow$ furniture	Abkhaz, Kashmiri, Meithei, Punjabi
Superordinate	$A + A \rightarrow N$	<i>Tibetan</i> big + small $\rightarrow$ size	Khmer, Mandarin
Superordinate	$A + A \rightarrow A$	<i>Vietnamese</i> strong + strong $\rightarrow$ healthy	
Superordinate	$V + V \rightarrow V$	<i>Meithei</i> eat + drink $\rightarrow$ dine	Vietnamese
Synthetic	$V + N \rightarrow N$	<i>French</i> carry + leaf $\rightarrow$ wallet	English, Fongbe, Hausa, Khmer, Meithei, Romance, Wolof, Yurakare
Conversion?	$A + V \rightarrow N$	<i>Tibetan</i> clear + show $\rightarrow$ demonstration	
Conversion?	$V + V \rightarrow N$	<i>Tibetan</i> say + say $\rightarrow$ speech	

Affixation vs. conversion: the resolution of conflicting patterns

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Linguistic economy assumes that no new forms are created if they do not add any new meanings to existing ones (see, for instance, discussions of blocking). It also assumes that, if two forms compete to express the same meaning and no other constraints apply, the one which strikes the better balance between formal economy and semantic distinctness will prevail.

Based on a list of 177 English adjectives derived in part from Dixon (1977) and Summers (1978), this paper examines how two conflicting patterns (-en suffixation and conversion) distribute themselves in the formation of corresponding deadjectival verbs. The adjectives are considered to be ‘basic’ ones in that they are not derived, they are largely Germanic or early Romance loans, they do not arise through conversion, and they are mostly monosyllabic. Evidence on the existence and lifespan of derived forms is taken from *The Oxford English Dictionary*. Focus is on how the two word-formation processes compete for the expression of largely the same meanings, how this competition may have affected the productivity of each type of form, and how this conflict is resolved.

In agreement with the principles of linguistic economy cited above, more verbs are lost in places where there is competition than where there is none. A tendency towards keeping the two forms can also be found, and also towards keeping -en suffixation when one of the two competing processes is lost. However, this does not explain how the competition is resolved. Three major variables, phonological, semantic and historical, are examined in the light of data of competing and non-competing deadjectival verbs for their influence on the outcomes found. Phonologically, the results agree with Marchand’s (1969: 272) description in that non-obstruent-final bases tend to derive their verbs by conversion, but they also disagree to some extent in that obstruent-final bases tend to form derived verbs both by conversion and by -en affixation. There is no hint of a preference for either of the processes on the basis of the semantic class of the adjective (Dixon 2005).

This paper is a case-study in what happens when word-formation processes compete for bases, and thus a case-study in the resolution of variation in the process of standardization (cf. Bauer 2006). The fundamental question is whether standardization, in reducing variability, leads to more or less predictable word-formation paradigms.

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So-called parasitic consonantal epenthesis and some aspects of diachronic morphology

Most atomistic theories of morphology would describe morphologically complex French words like *bijouterie* in terms of concatenation of atomistic categories smaller than the word such as *affix* (-erie) and *stem* (*bijou*) and consequently, non-etymological as well as non-phonological epenthetic consonants like (t) are labeled ‘parasitic’ (cf. Pagliano 2003) because they can be assigned to none of these units. In contrast, for the holistic theory called Whole Word Morphological (WWM) (elaborated in Ford et al. 1997), nothing can be parasitic in a word, because words have no internal structure. According to this approach, as soon as a speaker knows two pairs of words with the same formal and categorical differences as well as the same semantic relatedness, a morphological strategy may become part of his competence, which he can use, if the need arises, to form or to understand a new word or to retrieve a word that he may have forgotten temporarily.

1. /X/_n ↔ /Xtjə/_n ‘who deals with /X/’

lait [lɛ] ‘milk’ ↔ laitier [lɛtjə] ‘dairyman’

bijou [biʒu] ‘ornament’ ↔ bijoutier [biʒutjə] ‘jeweler’

Once we map complex words containing parasitic epenthetic consonants onto the appropriate strategies, it becomes clear that they are nothing but a theoretical illusion created by atomistic approaches of morphology. Diachronically, it can be noticed that (i) new strategies (e.g. (1)) emerge following phonological ([lɛt]>[lɛ]) change (cf. Fouché 1961) in the pair-mates of other strategies (e.g. (2)) and that (ii) a strategy can remain dormant for a certain period of time or (iii) can even die out if it lacks the minimum requirement of two pairs of words motivating it.

2. /XC/_n ↔ /XCjə/_n ‘who deals with /XC/’

lait [lɛt] ‘milk’ ↔ laitier [lɛtjə] ‘dairyman’ (12th)

fruit [frɥit] ‘fruit’ ↔ fruitier [frɥitjə] ‘fruit seller’ (1285)

I claim that (i) WWM gives a clearer idea of morphological changes in terms of death and birth of strategies following phonological change compare to other approaches which attempt to explain these changes in terms of the evolution of the morphemes and/or the change in their productivity and that (ii) it is possible to formulate a general theory of diachronic morphology within the theoretical approach of WWM.

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„Ich geh Aldi mit Bus“ – oder: Wohin steuert die Flexion der NP?

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Der Verlust der Flexionsmarkierungen ist ein seit vielen Jahrhunderten und auch gegenwärtig noch zu beobachtender Prozess. Dies war insofern unproblematisch, da die durch die Flexive angezeigte syntaktische Funktion durch die Formveränderungen der Artikel übernommen wurde. In den letzten Jahren scheint nun auch dieser Ort für die Markierung syntaktischer Relationen gefährdet zu sein, denn im mündlichen Sprachgebrauch vieler Jugendlicher (und das sind nicht nur Kinder mit Migrationshintergrund) ist bei der Produktion von NPs immer häufiger die Tilgung des Artikels zu beobachten. Zu diesem Phänomen wollen wir in unserem Vortrag einige theoretische Vorüberlegungen anstellen.

Offensichtlich erfolgt die Tilgung der Artikelform nicht ‚blind‘, sondern nach spezifischen Präferenzen und unter spezifischen Bedingungen. So kann der Artikel bei Präpositionalphrasen eher ausfallen als bei einfachen NPs und für diese scheint wiederum zu gelten, dass die NP in Objektfunktion eher artikellos bleiben kann als in Subjektfunktion. Schließlich ist das Ausmaß der Expansion einer NP nicht unerheblich für die Möglichkeit ihres artikellosen Auftretens. So deuten bisher eher zufällig erhobene Beobachtungsdaten in die Richtung, dass mit wachsender Komplexität, hier nur definiert über die Anzahl der attributiv gebrauchten Adjektive, die Wahrscheinlichkeit der Artikellosigkeit der NP zunimmt. Im Vortrag soll nun als Erklärungsansatz für dieses Tilgungsphänomen eine funktional motivierte Präferenzskala entwickelt werden.

Changes in function and inflection of indefinite articles in German

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German noun phrases containing an article realise gender and case inflection on that article. There seems to exist only a few exceptions to this rule, e.g. the indefinite article (*ein*) and, as well, the related negation article (*kein*) and the possessive articles (*mein*, *dein*, etc.) remain uninflected in nom.sg.msc. and nom./acc.sg.ntr. This is marked behaviour with respect to the dominating type of gender and case inflection. However, it will be argued, that there exists a markedness conflict between different structural parameters relevant for inflection and paradigm organization.

The behaviour of the indefinite article is motivated, and thus unmarked, with respect to the structural parameters case hierarchy, gender hierarchy, and paradigmatic preferences for the form of the respective articles. The development of nominal (or noun phrase) inflection in language history reveals that both the emergence of the indefinite article and of noun phrase inflection is part of a general reorganization of the formal means of nominal determination. The overt result of which is the tendency to analytical realisation of nominal inflection. As one non-overt result, central functions of nominal determination has become formally separated from inflection. Being functional heads the separate determiners attract other grammatical markings as well. However, the combined formal realisation of grammatical features does not involve a dependence of determination from inflection, thus the non-inflection of determiners in nom.sg.msc. and nom./acc.sg.ntr. is not marked with respect to nominal determination.

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Measuring inflectional economy

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Traditional approaches to morphology tend to treat inflectional systems not as unstructured sets of forms with shared stems or roots but as structured networks of elements. The interdependency of elements is, as Matthews (1991: 197) notes, 'the basis of exemplary paradigms' in the classical grammatical tradition. The same structure underlies the phenomenon of 'Paradigm Economy' in the sense of Carstairs (1983), as the interdependency of cells in an inflectional paradigm effectively constrains the number of inflection classes in a language. Although the exemplary patterns and leading forms of traditional descriptions bring out the structure of inflectional systems, traditional accounts are deficient -- or at least incomplete -- in a number of important respects. In particular, there is no method for measuring the implication structure of a set of forms or, no means of gauging the diagnostic value of specific forms within a set, and no generally accepted way even of identifying the leading forms of a system.

The problem of measuring this kind of inflectional structure can, however, be recast in a form that is susceptible to well-established techniques of analysis. The approach outlined in this paper proceeds from the observation that the relevant structure involves information that one form conveys about other forms. This information correlates with various of the **entropy** measures proposed initially in Shannon (1948). This paper first establishes baseline entropy values for an inflectional system by calculating the average entropy (i.e., uncertainty) of the realizations of each paradigm cell. The initial calculations are done on noun declensions in Russian, Serbian, Saami, Estonian and Nenets. So that the method applies to languages for which we lack frequency data, we provisionally assume that all realizations of a given paradigm cell are equiprobable. The diagnostic value of a cell C then correlates with the difference between the baseline entropy of a system and the **conditional entropy** of the system given knowledge of C. The implicational structure of the system likewise correlates with the average **mutual information** of pairs of cells in the system. The resulting measures identify the leading forms of a system as the realizations that minimize the entropy of the system. They also diagnose the anomaly of a fully suppletive class system or the pathologically 'uneconomical' classes of Carstairs (1983), since in neither type of system does knowledge about any form reduce the uncertainty of other forms. Furthermore, the application of standard information-theoretic techniques complements and helps to clarify previous implicative approaches to morphological analysis, such as Bochner (1993) and Finkel & Stump (2006).

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Paradigm shape is morphomic in Nepali

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Realizational morphology uses various devices to address discrepancies between feature sets and actual forms filling the paradigm: feature underspecification, rules of referral (Zwicky 1985), or a distinction between content paradigm and form paradigm (Stump 2006). We defend a radical version of the last proposal on the basis of Nepali conjugation.

Nepali conjugation exhibits 4 regular syncretisms (Baerman *et al.* 2005), illustrated in Fig. 1 with the long negative indicative present of *birsanu* ‘forget’. Whereas other syncretisms are local to a tense or inflection class, these hold for all verbs throughout the 16 synthetic tenses—even where alternate forms are available to fill a cell (e.g. FUT.POS.2.H2.PL *birsaulā* vs. *birselāu*), a pervasive situation in Nepali. In addition, natural classes of exponents do not coincide with natural classes of morphosyntactic features. We argue that this situation calls for an analysis where the 4 syncretisms are a consequence of the structure of the form paradigm, rather than the conspiracy of different rules of referral, which we would expect to be violated in at least some cases. We propose that content and form paradigms can be structured by distinct feature sets ; here the tense-internal form paradigm is structured by two morphomic features (Fig. 2). This provides a third instance of a morphomic phenomenon, in addition to stem alternations and inflection classes (Aronoff 1994).

	SG		
	F	M	PL
1	<i>birsād-in-ā</i>		<i>birsād-ayn-āũ</i>
2.H1	<i>birsād-in-as</i>	<i>birsād-ayn-as</i>	
2.H2	<i>birsād-in-au</i>		<i>birsād-ayn-au</i>
3.H1	<i>birsād-in-a</i>	<i>birsād-ayn-a</i>	
3.H2	<i>birsād-in-an</i>		<i>birsād-ayn-an</i>
H3	<i>birsanu-hūdayna</i>		

Fig. 1

	SG		
	F	M	PL
1	A-1		B-2
2.H1	A-3	B-3	
2.H2	A-4		B-4
3.H1	A-5	B-5	
3.H2	A-6		B-6
H3	7		

Fig. 2

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Abbreviations

FUT: futur

POS: positive polarity

NEG: negative polarity

SG: singular

PL: plural

M: masculine

F: feminine

H1: low grade honorific

H2: middle grade honorific

H3: high grade honorific

Sorani Kurdish person markers and the typology of agreement

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Person markers (PMs) vary along two dimensions: on the form dimension they range from full words through clitics to affixes; on the function dimension they vary from argument-filling pronouns to redundant agreement. We expect the two dimensions to be correlated: agreement is canonically affixal (Corbett, 2005), pronouns are canonically words. The Sorani Kurdish (SK) data show that the two dimensions are however strictly autonomous.

SK has two sets of PMs in addition to full pronouns. The first are clitics realizing on the first constituent of the VP (1,2,4), as endoclititics if that word is the verb (3). The second set are suffixes on the verb.

With intransitives and present transitives, affixal PMs realize subject agreement (1). In past transitives, they realize object agreement (clearly the remnant of an ergative system; Mackenzie, 1961) but are optional (2). Clitic PMs obligatorily mark subject agreement in the past transitive construction (2). In the present, they serve as pronominal objects (1,3)—they do not cooccur with an object NP (4). Thus there are three functions, each of the two sets of PMs can have two of these, and both can be agreement of the same type.

- (1) bâzîrgan-akân [_{VP} bo Sârâ=î da-kir-in]
merchant-DEF.PL for Sara=3SG IPFV-buy.PRS-3PL
'The merchants are buying it for Sara.'
- (2) bâzîrgan-aka [_{VP} asp-akân=î da-kirî(-n)]
merchant-DEF.SG horse-DEF.PL=3SG IPFV-buy.PST-3PL
'The merchant was buying the horses.'
- (3) bâzîrgan-akân [_{VP} da=î=kir-in]
merchant-DEF.PL IPFV=3SG=buy.PRS-3PL
'The merchants are buying it.'
- (4) *bâzîrgan-akân [_{VP} asp-aka=î da-kir-in]
merchant-DEF.PL horse-DEF.SG=3SG IPFV-buy.PST-3PL

We present an HPSG analysis incorporating an explicit realizational morphological component. Both clitics and affixal PMs are introduced by realization rules, but only clitics can be ordered outside of the word they originate from (Crysman 2002). The different functions of PMs correspond to different types of feature bundles to be realized.

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Construction-dependent morphology

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The insight of Construction Grammar that constructions form an essential level of description and generalization in the grammars of natural languages has a number of important implications for morphological theory and description. In my presentation, I will focus on the phenomenon of construction-dependent morphology: the proper morphological forms of words may be dependent on specific constructions in which these words occur. In the domain of contextual inflection it is quite obvious that the distribution of specific morphological forms of words depends on the syntactic context. What I want to show, however, is something different: that specific syntactic constructions require or allow specific inflectional or derivational affixes to be present in words of those constructions.

For example, in Dutch, we find expressions such as the following

Met ons twee-en / twee-tje-s / *met ons twee
With our two-PLUR / two-DIM-PL / *with us two.SG
'The two of us'

Wij drie-en / drie-tje-s / * drie
We three-PL / three-DIM-PL / * three.SG
'The three of us'

In these examples we see that cardinal number names must be pluralized (a case of inherent inflection) and optionally also diminutivized (a case of derivation). These morphological forms do not occur in other syntactic contexts. For instance, the following sentences in which the relevant suffixes occupy the same position in number words are ill-formed:

*Twee-en / Twee-tje-s komen later
Two-PL / Two-DIM-PL come later
'Two persons will come later'

Note that these plural forms are not cases of lexicalization since the relevant constructional scheme can be applied to all number names, and represents a productive pattern, as illustrated by *Met zijn dertig-en* 'The thirty of us'.

This implies that the syntax may require access to morphological properties of words (even derivational ones) for other purposes than agreement and government. Moreover, it shows that specific classes of complex words, for instance plural and diminutive forms of number names, are dependent for their usability and occurrence on specific syntactic constructions.

In my presentation I will discuss a number of such cases of this interwovenness of syntax and morphology. They provide strong evidence for the relevance of the notion 'construction', both at the syntactic and the morphological level.

What makes clitics tick?

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There are, broadly speaking, two types of analyses of clitics: the syntactic approaches (e.g. van Riemsdijk 1999) and the analyses of clitics as phrasal affixes (e.g. Miller 1992; Halpern 1995). The syntactic approach makes a distinction between clitic and affix, the former is a matter for syntax and the latter for morphology. In the phrasal affix approach, the two are more closely related. Indeed Anderson (1993, 1996, 2005) argues explicitly for the similarity between affixes and clitics. To him the difference is accounted for by different values for a constraint governing distribution by giving a value for attachment, or anchoring.

- (1) a. ANCHOR (FIRST/LAST) for clitics b. ANCHOR (HEAD) for affixes

An apparently obvious assumption would be that any element has its distribution governed either by a high ranking of (1a) **or** of (1b), elements are either head affixes or phrasal affixes. However, we present evidence that elements may be subject to a highly ranked (1a) **and** (1b) at the same time, so that the most optimal output is the one where both can be satisfied at the same time, i.e. when the head is at the edge of the phrase.

The element in question is the possessive *s* in Swedish and English. It is known as a clitic, since there are examples such as (2), where the *s* is attached on the right edge to a non-head constituent.

- (2) a. *the then president of America's daughter* b. *den dåvarande presidenten i USAs dotter*

However, such examples are rare. For instance, in the spoken part of the British National Corpus, only 0.04% of all uses of the possessive '*s* exemplify group genitives. Furthermore, there is evidence that speakers avoid this construction and use, for instance, the *of*-possessive. More relevant to us, there are avoidance strategies which involve using the possessive '*s*, but with head placement instead. This is the so called split possessive exemplified in (3), where noun phrase internal modification is extraposed to a position after the possesum.

- (3) a. *someone's baby in the audience*
b. *fofbollsupportarnas skrik som just sett sitt lag förlora*
 football supporters.s shouts who just seen their team lose

In (2), (1a) would appear to outrank (1b), but for the (3), the opposite order would hold. Both elements originate from a case suffix and it may be argued that this mixed behaviour is due to a state of historical flux; the element is developing from an affix to a clitic (hence degrammaticalising). However, the two construction types in (2) and (3) have coexisted for about 800 years, with apparently little change in the distribution. Instead we will argue that the generally assumed distinction between affix-clitic, or even affix-phrasal affix-clitic does not offer a sufficiently fine-grained system for characterising bound elements. Rather, the distinctions need to be discussed in terms of variation in separate dimensions.

On the basis of a database of possessive expressions built up from English and Swedish corpora, we define in detail the empirical facts relating to the structural and other properties which govern the expression of possessives generally and the distribution of the '*s* more specifically. We also propose constraints that would predict the correct outcome.

Re-redupli-ca-cation

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This paper reports on a cross-linguistic survey of reduplication. Thirty languages of genetically diverse origins and from geographically diverse locations were used as the sample for this study. This study is concerned primarily with productive reduplication and addresses several questions which are morphological in nature.

The cross-linguistic preference for suffixation in affixational morphology has been discussed by Cutler et al (1985). One of the central topics of this study is the relationship between reduplication and the suffixing preference. Reduplication is generally accepted to be a type of affixation where the shape and placement of the affix is specified, but the segmental material that makes up the affix is underspecified and must be copied from material in the base. The findings of this study show that prefixation is overwhelmingly preferred in reduplication. This is true for languages which are otherwise predominantly suffixing in their morphology as well as for languages which are predominantly prefixing. In addition to a prefixing preference infixation is also more prevalent in reduplication than elsewhere in morphology. This paper addresses the question of why there should be a preference for prefixing and a prevalence of infixation within reduplication, and explores the issue of how reduplication should be treated within the morphology.

The second key issue addressed in this study is the correlations between form and meaning in reduplicated constructions. This study considers whether or not there is a cross-linguistic preference for a one-to-one equivalence between reduplicated form and function; that is, whether it is the case that multiple different patterns of reduplication may express the same meaning, or whether multiple different meanings may be expressed by the same reduplication pattern(s).

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How to derive covert nominals: agentive/instrumental compounds in French

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Since the emergence of Distributional Morphology (DM), traditional analyses of complex nominals (derived nouns and synthetic compounds) involving argument structure have been analyzed (or reanalyzed) in different ways. These new analyses – Lexicalist, DM or “mixed” (e.g. Borer) – are successful to various degrees, and appear to be in complementary distribution as to their strengths and weaknesses.

I will put these various approaches to test through the analysis of agentive/instrumental compounds in French (e.g., *coupe-ongles* ‘nail clipper’; *ouvre-boîtes* ‘can opener’). These nominals have the following properties: 1) they are synthetic compounds; 2) their head is a noun, masculine, interpreted as the external argument of the verb from which it is derived, that is, an interpretation (and gender) that is identical to that of nominals derived through affixation of *-eur* (e.g. *coup-eur* ‘cutter’); 3) the head does not exhibit an overt morphological mark of its nominal property, as is the case in English, for instance (e.g. *coupe* rather than *coup-er* vs. *clipp-er*); 4) in the vast majority of cases, the head cannot be interpreted as a noun outside the compound without an overt nominal affix (e.g., *le tu-eur* but **le tue* “the killer”); 5) in contrast with other compounds, the mark of plural never appears on the head.

Specifically, I will address the following questions:

- In a Lexicalist approach (Lieber 1992, 2004), can we account for the properties of the head without resorting to the presence of a zero-affix, hence avoiding the conceptual problems linked to the existence of such affixes?
- In a DM approach (Arad 2003, Don 2004, Marantz 1997), can the distinctive behavior of forms like *coupe* or *tue* within and outside compounds be explained by means of the distinction between word-formation from words and word-formation from roots?
- In Borer’s approach (Borer 2003, 2005), can the distinction between lexical items that are category-neutral and those that are specified for category account for the properties of the head of these compounds?

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Formal Variation in French and Italian Prefixes: a Representation Model for Machine Translation

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In order to deal with constructed neologisms in machine translation systems, we propose to use bilingual Lexeme Formation Rules (LFR) that capture the construction process of neologisms in a contrastive manner. We present here a first attempt at a theoretical formalisation of these bilingual LFRs (for prefixation in French and Italian) and the various formal variations in the construction of neologisms that the translation perspective brings to light.

We propose to formalise LFRs as a quadruple (2 inputs and 2 outputs) as shown below in the rule for reiteration prefixation in Italian and French:

$$(IT) \text{LFR}_{\text{REITER}} [X]_V \rightarrow ri[X]_V \leftrightarrow (FR) \text{LFR}_{\text{REITER}} [X]_V \rightarrow re[X]_V$$

Obviously, this approach is confronted by numerous theoretical and practical difficulties. Many of them are monolingual issues, and will not be addressed here. Others difficulties comes from the variation amongst the multiple exponents found for some rules. Indeed, translation equivalence between two rules is often possible only if variation amongst multiple exponents is taken into account.

In this study, we first show how we set up translation equivalences between rules, using a semantic classification of prefixation that is used as a *tertium comparationis*, onto which prefixes from both languages are projected. From this projection, we obtain simple equivalences ($1 \leftrightarrow 1$) (like in the reiteration class, which contains two exponents ri_{IT} and re_{FR} and from which the rule shown above has been formalised) or more complex equivalences, like in the ‘multiplicity’ rule, where numerous exponents are possible ((IT) $\text{LFR}_{\text{MULTI}} [X]_{n|a} \rightarrow multi|pluri|poli[X]_{n|a} \leftrightarrow$ (FR) $\text{LFR}_{\text{MULTI}} [X]_{n|a} \rightarrow multi|pluri|poly [X]_{n|a}$). In these kinds of rules, monolingual restrictions might be put on the application of one prefix or another (for example, *poli* seems to prefer the base from the ‘technical domain’). But to translate neologisms (like the Italian *multidialezzale*), the automatic processing has to take into account the variation, and produce both possibilities (*multidialectal/pluridialectal*).

We will address the question of whether, for bilingual rules including multiple exponents, (such as (IT) $\text{LFR}(maxi/mega/meta) \leftrightarrow$ (FR) $\text{LFR}(maxi/méga/méta)$ or (IT) $\text{LFR}(iper/super) \leftrightarrow$ (FR) $\text{LFR}(hyper/super)$), the prefixes are interchangeable or whether the variation is conditioned by the monolingual restrictions that are imposed.

Part-Whole Relations in Slovak adjectival compounding.

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The aim of this paper is to examine the Part-Whole Relations [6], [7] as a semantic constraint involved in the Lexeme-Formation Rule (henceforth LFR) forming Slovak adjectival adjective (A)-noun (N) compounds (AN_A). Following the Lexeme-Based Approach [1], [4] in compounding phenomena [3], [5], this analysis is based on the *Slovak National Corpus* [8] and focuses on AN_A expressing a **Part-Whole Relation**, cf. (1) in sequences such as (1') where the AN_A modify a head noun (HdN).

(1) '[HdN_{Whole}] whose / which N_{Part} is A'

(1') [motýl_{HdN}] VEJÁROVÝ_A, KRÍDLO_N →_{compA_N} VEJÁROVOKRÍDLY_A fan-winged
butterfly fan-ový_A wing_N fan-winged butterfly

I will show which semantic constraints are imposed by this LFR both to the nominal component, i.e. N_{Part} and to the HdN_{Whole}. Among others, we will see that the HdN_{Whole} may denote either a [+NATURAL] or a [-NATURAL] concrete entity whereas the N_{Part} can never refer to an artefact excepted when the N_{Part}-HdN_{Whole} relation is interpreted as a **Natural Attachment** [2], cf. (2)-(2').

(2) '[HdN_{Whole}] (Ground), location entity / N_{Part} (Figure), localized entity'

(2') [hrebeň_{HdN}] ŠIROKÝ_A, ZUB_N →_{compA_N} ŠIROKOZUBÝ_A large-toothed
comb large_A tooth_N large-toothed comb

Considering six Part-Whole Relations classes defined by [6] (following [7]), I will point out that two of them (i.e. 'member/collection', 'sub-collection/collection') never occur in AN_A-HdN sequences. We will also see that meeting both Part-Whole Relation and Natural Attachment criteria is not a sufficient condition for a AN_A-HdN sequence to be well-formed (3).

(3) [postel_{HdN}] *MÄKKÝ_A, MATRAC_N →_{compA_N} MÄKKOMATRACÝ_A soft-mattressed
bed soft_A mattress_N soft-mattressed bed

It seems that the **semantic type criterion** [+/-NATURAL] of N_{Part} in this LFR is more significant than the identification of conceptual proximity between N_{Part}-HdN_{Whole} (3').

(3') [postel_{HdN}] KRÁTKY_A, NOHA_N →_{compA_N} KRÁTKONOHÝ_A short-legged
bed short_A leg_N short-legged bed

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Prolegomena to a typology of periphrasis

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Periphrasis straddles the boundary between syntax and morphology, and so creates analytical and theoretical problems. These concern the nature of the word, the interaction between syntax and morphology, and the possible sizes and shapes of inflectional paradigms. Progress has been hampered because the theoretical devices available have been inadequate and because the range of data considered has been narrow. We are therefore undertaking a typological survey, adopting a ‘canonical’ approach in which we specify the different dimensions along which concrete instances of periphrasis can be classified as more or less canonical.

Periphrastic constructions are common, yet relatively little is known about their typological range. Following Haspelmath (2000), our canonical approach encompasses not just verbs but also comparable phenomena involving other parts of speech (‘analytic’ forms in some traditions). Thus we include less familiar examples, such as this one, showing periphrastic expression of number on nouns. We give just two of seven cases:

Nenets *ti* ‘reindeer’ (Ackerman 2000:3)

	SINGULAR	DUAL	PLURAL
NOMINATIVE	ti	tex ^o h	tiq
DATIVE	ten ^o h	tex^oh nyah	tex ^o q

Number and case are expressed synthetically for most cells of the paradigm, except for certain cells in the dual (represented here by the dative).

We plan a careful examination of four key languages, Archi, Biniṅ Gun-Wok, Nenets and Sanskrit, each of which has significant and different instances of periphrasis. Based on this survey, we will establish:

- the set of criteria for periphrasis; this is to define the space across which to calibrate any attested periphrastic construction. We shall start from the definitions in Ackerman & Stump (2004: 125-147).
- the typological range of periphrasis, taking into account both the constructions and how they interact with language-specific morphological and syntactic rules
- the diachronic stages that the syntactic phrase goes through on its way to becoming part of the morphological paradigm, and the conditions that permit this.

Periphrasis can be understood in terms of a configuration of properties of canonical inflectional morphology (e.g. obligatoriness) and canonical syntax (e.g. independence of expression), as in our Nenets example. The canonical approach with its inclusive orientation allows us to calibrate and clarify the differing intuitions of typologists, and place periphrasis within the larger space of morphology-syntax interaction.

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On the origin, expansion and productivity of a Romanian conjugational (sub-) class

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As in certain other Romance languages, the Romanian 4th conjugation (verbs in **-i**) is heavily populated by verbs that show a stem extension from the Latin inchoative affix **ISC/ESC** in some forms (e.g. **doresc** 'want_{1SG}', **iubesti** 'love_{2SG}', etc.). While most Romanian **i**-verbs follow this conjugational pattern, not all do. For instance, **a dormi** 'sleep_{INF}' has **dorm** 'sleep_{1SG}', rather than ***dormesc**. In most descriptions of the Romanian conjugational class system, **i**-verbs are broken down into two subclasses, one with the reflex of the inchoative affix ([+sc]), and the other without it ([-sc]). Even so, many verbs fluctuate between the subclasses dialectally.

While verbs in the [-sc] subclass are predominantly core vocabulary inherited from Latin, the [+sc] subclass is mainly populated by two groups. Many are verbs borrowed from other Balkan languages, mainly S. Slavic (e.g. Rmn. **iubesc** 'love_{1SG}' < OCS **любити** 'love_{INF}'). The other main group are denominative verbs formed from nouns/adjectives regardless of origin (e.g. Rmn. **îmbolnăvesc** 'make sick_{1SG}' < Bg. **болнав** 'sick_{ADJ}', Rmn. **numesc** 'name_{1SG}' < Lat. **NOMEN** 'name_{NOUN}'). This suggests that a similar derivational strategy is responsible for the assimilation of foreign items and the formation of denominative verbs.

However, not all Balkan borrowings fall into the [+sc] class of the 4th conjugation, as a few fall into the [-sc] class or into other conjugational classes altogether. In addition, there is variation across dialects to which class some of these verbs were incorporated into. This variation continues into the contemporary language as well, as shown by forms of recent loans such as **bloguiesc/bloghez** < Eng. **blog**.

The variation is indicative of different models being employed in the assimilation process inasmuch as the expansion of this class is essentially a result of analogical change. When words are borrowed, they must be assimilated to previously established patterns in the recipient language. Moreover, the same analogical processes active in the past remain active, so that as Labov (1994) suggests, we can learn about developments in the past by examining the present.

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The Latin *-alis/aris* allomorphy revisited

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The Latin *-alis/aris* allomorphy figures prominently in the literature (Steriade 1987, 1995, Kenstowicz 1993 etc.) and has been the object of detailed investigation (Watkins 1970, Dressler 1971, Hurch 1991), though the nature of the patterns the data display and their interaction with phonotactics are not fully appreciated. A corpus-based analysis (CLCLT5–Library of Latin Texts) of the distribution of liquids reveals asymmetries between **r** and **l** (some of which have been known earlier) and also important differences in repetition patterns which have been only sporadically studied so far.

For **l** the constraint appears to be that within a word, its occurrences must be separated by at least one non-coronal consonant; the productive diminutive formation of the type *puellula* 'little girl' is a saliently exceptional pattern. The case of the *-alis/aris* dissimilation falls under this constraint: contrary to the commonly held view, liquid dissimilation is not only blocked by an intervening **r** but also by any intervening non-coronal consonant (*glacialis*, *pluvialis*). The distance between the two liquids is also relevant in that more than two morae are required for blocking to be effective.

The case of **r** is different in many respects. In final syllables only **rVr**, in non-final syllables only **rVVr** occurs. While this receives a partial diachronic explanation in a well known prehistoric sound change, the interesting and non-obvious fact is that Latin word forms generally conform morphologically to this pattern even where this is not warranted diachronically, i.e. the prefix *re-* never attaches to stems beginning with *r*, and *r*-initial verbs never display reduplication, thus no non-final *rěr-* sequences emerge even where they in principle could. The exceptions to this generalisation are explained with reference to two different morphophonological factors.

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French ethnic property nouns: a case of base variation

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The present study (see [2]) is about French *-ité* suffixed nouns expressing the property both of those who belong to a place (ITALIANITÉ), and of the place itself (FRANCITÉ). We call these nouns « ethnic property nouns » (**Nep**).

1. Issue: There are two observed ways for a speaker to coin a **Nep**.

– First, applying the general French *-ité* Lexeme Formation Rule ([3]) to an **ethnic adjective** (**A_E**) input: e.g. ITALIANITÉ,

– Second, applying *-ité* directly to the **toponym** (**N_T**), e.g. PORTUGALITÉ.

Both strategies can be used to produce the same semantic output. But when it happens, it is with different frequencies (e.g. BELGITÉ: 31 vs. BELGICITÉ: 142).

The question we explore is whether it is possible to predict the most likely form of **Nep**.

2. Data Description and typology: We have collected 212 **Nep** on the Internet related to 145 **N_T** out of a set of 412 candidate **Nep**, coined from either **N_T** or their related **A_E**. These **Nep** are equally distributed, as far as their base is concerned, among **N_T** (45), **A_E** (46) and both (38). In 16 cases, we can not decide whether the input is a **N_T** or an **A_E**.

We consider the application of Lexeme Formation Rules as the default case. Our data show that its instantiation can be favoured or put in a disadvantage by four output-applied trends, which can be expressed in terms of prevention (a-b) or preferences (c-d): (a) very strong, if not absolute, prevention of contiguous sequences with identical or similar phonemes (dissimilative constraint): this explains the lack of /njanite/ ending **Nep** (IRANIANITÉ: 0 occ). (b) strong prevention of *-aisité* and *-oisité* final sequences. (c) as for Spanish (F. Rainer, pers. com), preference for well represented final sequences in the French attested lexicon: /<C_{ons}>anite/ (e.g. EGYPTIANITÉ), /alite/ (e.g. PORTUGALITÉ). (d) preference for quadrisyllabic outputs:[4]. For an output, two or more of these four trends can be cumulated.

3. Theoretical implications: Two main reasons make **Nep** theoretically interesting:

(1) Procedural (rules) and declarative (output constraints) approaches define orthogonal though not mutually exclusive LF principles. Our data prove that these approaches should not be opposed to each other, since the decision to assign **Nep** construction exclusively to rules or to a constrained-based system ('Output-to-Output faithfulness principle': [1]), leads to a lack of generality.

(2) these two competing ways of forming French *-ité* **Nep** show that, from the speaker viewpoint, an **A_E** and its corresponding **N_T** are semantically equivalent for *-ité* suffixation.

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List of abbreviations

Ne.pr : Ethnic property noun	A_E : Ethnic Adjective	N_T : Toponym
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Maltese nouns and adjectives: (dis)agreements

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I. Data

Aquilina (1965), Borg & Azzopardi-Alexander (1997) and Sutcliffe (1936) distinguish three forms in the paradigm of the adjective. The masculine singular form is unmarked and used as the base to form the feminine singular and the plural: sound, broken or invariable. There is no distinction of gender in plural, adjectives possess a unique form.

PLURAL	EX	SINGULAR		PLURAL	TRANSLATION
		Masculine	Feminine		
Sound	(1)	baħnan	baħnana	baħnanin	stupid
Broken	(2)	ħafif	ħafifa	ħfief	light
Invariable	(3)	vjola	vjola	vjola	violet

II. Problems

- The formation of the feminine singular form is always regular but it is not the case for the plural form. So we use a two-slot stem-space following Bonami & Boyé (2003) and Pirelli & Battista (2000) to capture the broken plurals.
- The form of the plural of adjectives is not always required with a plural noun. Few nouns in plural accept the feminine singular form of the adjective.

(4.a)	irġiel tajba	(4.b)	irġiel tajbin
	MEN GOOD-F/S		MEN GOOD-PL
	‘good men’		‘good men’

Maltese native speakers do not consider *tajba* as a plural form.

It means that few adjectives possess two forms for the plural. The presence of two forms in the same slot is problematic and the non-contrast between one of the plural forms and the feminine singular form is problematic for the Word and Paradigm theory.

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Morphological Variation and Change in the Rigveda: -au vs. ā

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Allomorphy between -ā and -au in duals and in *i*-stem locatives in the Sanskrit of the Rigveda poses an interesting challenge for linguists and Sanskritists due to the multi-faceted nature of the principles governing the variation. Both traditional and more recent accounts of this alternation (e.g. Wackernagel 1930, Malzahn 1999) present a generalized, purportedly original distribution: -ā appears preconsonantly and -āv prevocally (with dual -ā vs. locative -au prepausally). Cases that do not fit into this generalization, such as preconsonantal -au, are attributed to chronological (i.e. date of a hymn's composition) or phonological factors.

An exhaustive compilation, however, of each such form in the Rigveda (over 3,000 total) reveals that while a hymn's date of composition can be relevant (i.e. more preconsonantal -au forms appear in later hymns; see e.g. Lanman 1880), its effects are not so simple to discern: many examples of preconsonantal -au are found in earlier hymns, and even later hymns usually have internal variation between -au and -ā. Phonological explanations are also found to be lacking.

No simple generalization can be proposed that can account for the data in this case: only in the interaction of various factors on certain forms in particular contexts can the explanation be found. The distribution of the forms in the Rigveda reflects variation and change taking place in the living language at the time of composition, and points in the direction of the change in these forms seen in the later language. I explore this situation from both the synchronic and diachronic perspectives, showing how the distribution of the -ā and -au allomorphs is affected by various phonological, metrical, lexical, and poetic factors, and showing how these same factors together with the synchronic patterns of variation resulted in the analogical extension of the variant -au (with complete loss of -ā) in both morphological categories by the time of Classical Sanskrit.

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On varying degrees of compositionality in word formation

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We present an analysis of the variation in word formation processes in the Flemish dialects, used in the northern Dutch-speaking part of Belgium. This analysis is inspired by Wray & Grace's (2007) hypothesis on degrees of compositionality. Developing Thurston's (1994) observation that a community's communication can be predominantly *esoteric* (inward-facing) or *exoteric* (outward-facing), they hypothesize that languages that are customarily used esoterically will tend towards lower degrees of compositionality.

Having analysed the morphological data in the dictionary of Flemish dialects (Ryckbeoer 1993), we can make three claims:

- (1) In particular lexical domains of the Flemish dialects the less-compositional word formation types do indeed outnumber the more compositional ones. In order to found this claim we will first distinguish between different degrees of compositionality in word formation, illustrating the different types of derivation and composition involved with relevant data from the Flemish dialect lexicon in the traditional domain of farming.
- (2) In particular lexical domains concepts get up to a hundred different names in the different dialects, illustrating a wide variety of more and less compositional word formation types.
- (3) Both low degrees of compositionality in word formation (claim 1) and high degrees of morphological dialectal variation in particular lexical domains (claim 2) can be related to the *exclusively oral* and the *predominantly esoteric character of dialectal communication in these lexical domains*.

We will relate our own findings to the data and hypotheses in Stekauer et al. (2005), who study creativity in English word formation from a sociolinguistic perspective. They examine the influence of education, profession and language background on the preference for either explicitness of expression or economy of expression in selecting more or less compositional word formation processes.

Thus confronting our own data on Flemish dialectal variation in word formation with some recent findings and/or hypotheses on both early human language and a modern standard language (English), we hope to contribute to the elaboration of a general linguistic theory on degrees of compositionality in word formation.

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On (im)possible morphological variation

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We argue that morphological variation, while being brought about by experience, is nevertheless subject to the constrained imposed the operations of the language faculty, and is enabled by changes in the properties of formal features.

We focus on the form of compounds, and in particular the recursivity, and the linear ordering their constituents. First, we argue that there is no possible variation regarding the recursivity of compounds. All the expressions generated by the grammar are theoretically infinite, and there is no limitation on recursivity other than the one imposed by human working memory. Second, we argue that possible morphological variation comes from a change affecting the content of the formal features. This is a consequence of the properties of the feature system of the language faculty, and its use by the language engine, which includes interpretable and uninterpretable (active) features.

1. Recursivity. According to Chomsky (2004), Hauser, Chomsky, and Fitch (2002), recursivity is the central property of the language faculty, and is part of the genetic endowment that makes humans capable of developing a grammar on the basis of experience. We argue that morphological variation cannot affect the recursive property of the operations of the grammar.

(1) $\text{Merge}(\alpha, \beta) > \text{LB}(\alpha, \beta)$

Thus there is no possible parameter that would differentiate languages with recursive compounds and languages without recursive compounds (but see Roeper, Snyder and Hiramatsu (2000) for a different view). We consider the properties of recursive operations, and we show that recursive constituents are part of English as well as French compounds, e.g., *alpha-amino-acids* (En), *acides alpha-aminés* (Fr). We discuss the directionality of the recursion (right-branching vs. left-branching) and attribute the difference to independent properties of the languages under consideration, along the lines of Di Sciullo (2005). Moreover, we consider the properties of morphological and syntactic recursivity and will derive the differences between English and French compounds from the properties of context-free vs. context sensitive grammars.

2. Formal features. We argue that possible morphological variation stems from the properties of the features available in the language faculty, and their role in the derivations. We propose that the variation in the order of the constituents in compounds is a consequence of the change in the properties of formal features, such that some uninterpretable feature becomes interpretable. The reason why the logical object of a verb precedes the verbal root in English verbal compounds is that the c-selectional feature of the verb, i.e., $[\text{uD}]$ is no longer an uninterpretable feature, i.e., $[\text{D}]$, (2), thus, it does not require checking under Agree. Consequently, the logical object does not occupy the canonical object position, e.g., *car-wash*, *ball-through*, *book-shop*. This is not the case in French, where the c-selectional feature of the deverbal compounds are checked in the canonical object position, e.g., *un lave-auto*, *un lance-pierre*, *un pare-brise*, etc.

(2) $[\text{uD}] > [\text{D}]$

(3) $[\text{D}] > [\text{uD}]$

If Deprez (2007) is right in arguing that the variation in the linear order of the constituents in DPs in French based Creoles is due to a change in the properties of the functional features such that certain interpretable features become uninterpretable, (3), morphological and syntactic change could be viewed as complementary.

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Affix borrowing or native developments? Dutch *-matig*

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It is well-known that intensive language contact can lead to affix borrowing. Yet it can be difficult to distinguish between borrowing and autonomous processes:

When large numbers of words or phrases are borrowed (...) or a particular class is borrowed (...), the question naturally arises to what extent the morphosyntax and semantics/pragmatics of the source language carry over into and affect native developments (Brinton & Traugott 2005:159).

I will discuss this for affix borrowing with a case study of the Dutch suffix *-matig*, which is a productive means for deriving adjectives from non-adjectival bases as in (1) (attributive adjective):

- (1) Op zoek naar manieren om werkplekken efficiënter te beheren? E-Quest heeft een **software-matige** oplossing voor u.
Looking for ways to manage workstations more efficiently? E-Quest has a software-wise solution for you.

Dutch grammar tends to reject derivation with *-matig* since it is felt to be “un-Dutch” (ANS): it is considered to be copied from word formation with *-mäßig* in German. Historically there are indeed indications that *-matig* was borrowed from German. Synchronically, *-matig* and *-mäßig* have similar semantic, morphological and functional properties. Particularly striking is their almost free combinability with substantival bases and their ability to express several fairly abstract semantic relations. Inghult (1975) has shown that *-mäßig* recently developed a ‘limitative’ meaning, which has become very productive and often combines with adverbial function; we find similar patterns in Dutch:

- (2) **Zangmatig** zijn ze erg goed, maar hebben niet de uitstraling van bijvoorbeeld de Spice Girls.
Singing-wise they’re very good, but they don’t have the charisma of for example the Spice Girls.

Do these similarities automatically imply that German sets the example for Dutch, as prescriptive grammar seems to suggest? I present some arguments which support the view that the linguistic properties of Dutch *-matig* could result from native developments as well. I refer to the Swedish suffix *-mässig*, which, according to Söderbergh (1964), has similar properties. This could be an indication for converging developments in the Germanic languages.

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Productivity measurement of English root compounding

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Morphological productivity measurement is an area of study which stands as a very attractive notion of word-formation, however most of its alternatives focus only on the productivity of affixation, ignoring other word-formation patterns of English. An especially flagrant case is that of compounding. Widely taken as a very profitable process in English, there have been no proposals to compute its productivity up to now. The present study stems from this necessity and focuses on root compounds of the type NN.

A natural obstacle is the formal structure of these units because, while affixes bear specific meanings, compounds have an identical structural makeup which prevents identifying semantic traits so easily. To overcome this, one possibility is focusing on meaning, which is common to all word-formation processes regardless of their formal makeup.

Thus, a corpus of compounds is analyzed using Levi's (1978) set of predicates and Huddleston's (2002) semantic roles. The former allows referring, for example, not to NN compounds in general but to NN-IN or NN-USE predicates, depending on the relevant predicate in each case, while with the latter NN compounds can be separated depending on their different semantic role (Agent, Instrument, Factive, Source, Goal, Location...).

In this paper, a sample of the labels Agent, Instrument and Spatial is picked up and analyzed for possibilities in measuring the productivity of NN compounds. This allows making generalizations as well as detecting problems and peculiarities of each set of units. The following are some examples from each of the labels:

Agent: *art historian, art student, art boss, athletics official, auto freak, bathroom specialist, beauty queen, blood relative, book people, business community, cab firm...*

Instrument: *aerobics mat, air defence weapon, air unit, airbrush, aircraft, alarm bell, ale tap, algae trap, amnesty product, anaesthetic machine, arm band, aromatherapy oil, art case...*

Spatial: *accident department, account house, air base, air museum, aircraft museum, airfield airport, airstrip, army war college, art gallery, automobile museum, business district...*

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Change and variation in morphonotactics

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Morphonotactics is the area of interaction between morphotactics and phonotactics (cf. Dressler and Dziubalska-Kołaczyk 2006). Phonotactic preferences hold for the basic forms of monomorphemic words; the less respected the preferences are, the more marked phonotactics sequences arise. Morphonotactic sequences, which are due to morphological operations of inflection or word-formation, are much more likely to be marked. Among the morphological processes creating morphonotactic sequences there is a preference for transparent morphological concatenation. In this contribution we limit ourselves to morphonotactic consonant clusters.

In Beats-and-Binding phonotactics (cf. Dziubalska-Kołaczyk 2002, Dziubalska-Kołaczyk & Krynicki 2007) the phonotactic preferences specify the universally required relationships between net auditory distances within clusters which guarantee, if respected, preservation of clusters. Clusters, in order to survive, must be sustained by some force counteracting the overwhelming tendency to reduce towards CV's. This force is a perceptual contrast modeled by Net Auditory Distance (NAD). NAD is the sum of the distances between segments with respect to manner of articulation (MOA), place of articulation (POA) and laryngeal activity (Lx).

Our claim about the markedness of morphonotactic sequences will be tested against the data from prefixation in Polish and compounding in Lithuanian.

The diachronic genesis of morphonotactic consonant clusters presupposes, on the one hand, a sufficient amount of morphological complexity. For example, morphonotactic clusters originating from non-concatenative operations such as morphological vowel deletion often occur in strongly inflecting but neither in weakly inflecting nor agglutinating languages. On the other hand, morphonotactic clusters arise only in languages with complex phonotactics. This will be illustrated with the history of (mor)phonotactics from reconstructed Proto-Indo-European up to modern Balto-Slavic languages.

This long history can also exemplify the demise of morphonotactics due to degrammaticalization, loss of morphological boundaries and alternations (Slovak *mach*, Gen. *mach-u* 'moss' vs. more conservative Polish *mech*, Gen. *mch-u*), analogy (e.g. *spotk+ł* → *spotk+a+ł* 'he met'), loss of other morphological rules (e.g. Lith. Gen. *šun-s* > *šun-io*, and also due to general phonotactic simplifications. The latter may start with variation in casual speech, e.g. in Polish *szed+ł* → [šet] 'he was going', gen.pl. *will* → [vil] 'villas'. All this fits to our modeling of morphonotactics within morphonology as area of interaction between morphology and phonology.

Morphological rules in the construction of the French names of inhabitants

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The French names of inhabitants (*Paris* → *Parisien* ‘Parisian’, *Toulouse* → *Toulousain*, *Nantes* → *Nantais*) are mostly constructed by derivation of the toponyms. There is a range of suffixes mainly used for detoponomastic adjectives, like *-ois*, *-ais*, *-ien*, *-ain*, These suffixes are not freely available for derivation to names of inhabitants (called *gentilés* in French), but their selection is determined by morphological rules. In addition, regional preferences for certain suffixes can be detected, e.g. the suffix *-ais* that is primarily found in the names of the inhabitants of places situated in western France. With regard to morphological determination, our contribution shows that the selection of a suffix doesn’t depend on the formal constitution of the toponym, but on the form of the derivational base (we call *radical*) chosen for the suffixation: e.g. the place name *Acigné* could form the non-attested gentilé **Acignéen* with the suffix *-éen* because of the ending *-é*, but as the base ending on *-ol* is chosen for the official *Acignolais*, the suffix *-ais* is determined. The construction of the derivational base depends on the applied morphological processes like epenthesis, allomorphy, or suppletion, as can be seen in several gentilés ending in *-éen*, *-ésien* or *-olais* built on place names ending in *-é*. As the rules don’t determine a single suffix, but rather restrict the selection to some of them and refuse the adjunction of another range of suffixes, the output in the building of French gentilés is related in a first step to the possible constructions of the derivational base and in a second step to the probable determinations of a suffix in order to get a gentilé with a harmonic structure. As a result, morphological variation can be detected in the construction of names of inhabitants that are not only due to the geographical preponderance.

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Case reduction in Norwegian (and Swedish)

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Old Norse nouns inflected for four different cases (nominative, accusative, dative and genitive). Most modern Norwegian (and Swedish) dialects have no case inflection. The obvious question becomes: Which form persists when four becomes one – which Old Norse case form serves as the formal basis for the modern form? From Mayerthaler's (1981) version of Natural Morphology, one would expect the nominative; it is the 'unmarked' case. In many West Norwegian dialects, this prediction is borne out: The nominative has indeed been generalised. Many East Norwegian dialects, however, have generalised non-nominative forms. Furthermore, many Swedish dialects display a "split" that is correlated with the animacy hierarchy: Nouns denoting animates have a form that goes back on the nominative, while other nouns have a form going back on the non-nominative.

Mayerthaler argues that, in cases where the original coding pattern is 'counter-iconic' (i.e., a semantic 'more' corresponds to a formal 'less'), there is a markedness reversal, and that this will influence the diachronic development. Norwegian provides a useful testing-ground for this claim. The reason is that some Old Norse nouns display what Mayerthaler would call counter-iconic marking; the nominative is characterised by a suffix, the accusative has none. Other Old Norse nouns do not display such 'counter-iconic marking'. The evidence indicates that these two groups do not really develop differently.

Mayerthaler's use of 'markedness reversal' and his claims about the nominative have been criticised before, for Romance (Carstairs-McCarthy 1992), but there are several reasons to go over this ground again. There is a value in a broader empirical perspective. Furthermore, there is a debate over the concept of 'markedness' (Haspelmath 2006, Gaeta 2006), and the evidence may be of interest here.

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Variation and change in the paradigm of Spanish object agreement markers

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A little studied aspect of Spanish morphosyntax is the remarkable dialectal variation exhibited in the paradigm of object agreement markers in non-standard rural varieties. Standard second person plural object marker *os* is realized /bos/~bus/, /sos/~sus/, /us/, /los/, /tos/ and /se/ in rural varieties of Peninsular Spanish while standard first person plural object marker *nos* is realized /mos/~mus/ and /los/. Because Spanish dialectological studies have focused traditionally on phonetic and lexical phenomena while devoting little attention to morphosyntactic variation, these non-standard uses have gone largely unnoticed in the literature. The few studies that have noted the existence of these forms have attributed them to analogy with some other form in the paradigm. However, no evidence has been provided to substantiate such claims and, what is more important, no functional explanation has been provided to account for why *os* and, to a lesser degree, *nos* are subjected to such frequent substitutions.

In order to investigate the realizations of *os* and *nos* in rural Spanish this study uses data from the Linguistic Atlas of the Iberian Peninsula (ALPI), online at www.alpi.ca, which contains information on these forms gathered in approximately 280 Spanish speaking localities. Whenever possible, the data from ALPI is complemented with more recent linguistic atlases, and with historical corpora.

The data illustrates well-documented universal tendencies regarding the evolution of grammatical forms in inflectional paradigms: these include not only analogical changes but also paradigm levelling, phonetic erosion, and preference for universally unmarked coronal sounds. Likewise, /los/ that substitutes for *os* or *nos* allows a paradigmatic reduction to one single marker for all the meanings conveyed by *os*, *nos* and *los* in standard Spanish. Merges of this sort are consonant with the general economic motivation of the changes studied, as homonymy is a common feature that contributes to economy and efficiency in inflexional paradigms. Furthermore, the direction of these merges seems to be dictated by universal tendencies according to which marked elements in inflectional paradigms are relatively more vulnerable to undergo coalescence with other forms.

Selectional constraints in Modern Greek suffixation: the case of verb-forming suffixes

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This paper focuses on the selectional constraints of Modern Greek verb-forming suffixes. Modern Greek has six main verb-forming suffixes (attached to nominal or adjectival bases): a) *-ono*, e.g. *pliji* ‘wound’ → *pligh-ono* ‘wound’, b) *-eno*, e.g. *pah-is* ‘fat’ → *pah-eno* ‘fatten’, c) *-evo*, e.g. *kaliteros* ‘better’ → *kaliter-evo* ‘become better’, d) *-izo*, e.g. *psifos* ‘vote’ → *psif-izo* ‘to vote’, e) *-azo*, e.g. *steji* ‘roof’ → *stegh-azo* ‘to roof’, and f) *-aro*, e.g. *control* ‘control’ → *control-aro* ‘control’. We will argue that the selectional constraints of these verb-forming suffixes are related to the phonological, morphological, semantic and etymological properties of their base (see also Plag 1999, 2003, Hay, J. & I. Plag 2004, Lieber 2004 for English). For example, a) *-eno* is mostly attached to monosyllabic adjectival bases in *-is*, *-ia*, *-i* (second inflection class), e.g. *pah-is* ‘fat’ → *pah-eno* ‘to fatten’, b) when two rival suffixes can be attached to the same base, e.g. *glik-os* ‘sweet’ → *glic-izo* ‘become sweetish’ & *glic-eno* ‘sweeten’, the basic selectional criterion seems to be both the meanings of the suffix and the base, c) *-ono*, *-eno*, *-evo*, *-izo*, *-azo* and *-aro* avoid attaching to bases derived with other suffixes, whereas final combining forms like *-poio* (e.g. *gramma-tikos* ‘grammatical’ → *grammat-ik-o-poio* ‘grammaticalize’) attach to both derived and simplex bases. We will also discuss conversion in relation to the overt suffixes (e.g. *tripa* ‘hole’ > *tripo* ‘make a hole’) and final combining forms. We will try to examine these constraints according to their degree of obligatoriness and/or priority. The morphophonological features of the suffixes will be handled along the lines of Morphological Feature Theory (Shieber 1986, Steele 1995, Ralli 2005 among others), Subcategorization Frame Theory (Aronoff 1976, Lieber 1980), System-Dependency Theory (Dressler 2003) and Optimality Theory (see Prince and Smolensky 1993). Data will be supported by an offline test, where native speakers will be asked to produce coinages using verbal suffixes.

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The invisible hand of grammaticalization: West-Germanic substitutive infinitive and the prefix *ge-*

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How far can grammaticalization have an impact on morphology? Besides superficial effects like those shown by so-called ‘trapped’ morphemes (cf. Harris & Faarlund 2006), much deeper consequences are revealed by a mystery tour through the West-Germanic verb morphology centered on the so-called substitutive infinitive, which shows up in the perfect paradigm when the latter is embedded into a verb complex: cf. the German example *Ulli hat die Morphologie-Tagung nicht besser organisieren_{INF} können_{INF}* ‘Ulli couldn’t organize better the morphology conference’. Here, a parasitic infinitival form takes the place of the expected past participle, which is also attested when there is no such a verb complex: *Ulli hat es so gewollt_{PTC}* ‘Ulli wanted it so’. Independently of the intricacies of the exceptional syntactic order shown by the verb complex when it occurs in a subordinate clause (cf. Schmid 2005 for a survey), for which a separate story must be told (cf. Gaeta 2005, 2006), the hypothesis will be laid down here that the reason for this mysterious case of form-meaning mismatch (cf. Gaeta in press) must be sought in the grammaticalization of the Germanic prefix **ga-* which intrinsically displayed a [+ bounded] semantic feature, as shown by its derivational behavior: Goth. *ga-nisan* ‘to be cured’, OE *ge-āscian* ‘gain by asking’, etc. (cf. Lowe 1972: 224). The prefix provided a verb past participle with an intrinsic telicity, as shown by its absence in (and late spread to) the past participles of already telic verbs such as OHG *funtan* ‘found’, *troffan* ‘met’, *wortan* ‘become’, etc., continued in the dialectal forms *fun*, *trof*, *wört* (cf. Schirmunski 1962: 517). This is the reason why it could not be combined with a past participle when the latter governed a further verb: the only condition for this to be possible was simultaneity between the two events (cf. MHG *eines tages vruo sach man dort rîten/*geriten* ‘one day one saw (him) riding there early’). This rendered the first predicate incompatible with boundedness because its second argument-time window was saturated by a predicate carrying an explicit [- bounded] feature (cf. Klein 2000 for details). Therefore, the intrinsically telic *ge*-participle was not allowed and the arguably default form, the infinitive, set in to respond to the strive towards the paradigmatic diffusion of the perfect construction in the late Middle Age. In this light, the long-wave effect of grammaticalization can be made responsible for the anomaly preserved until today in all West-Germanic dialects, in which *ge-* was grammaticalized as an inflectional marker.

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Variation of stressed vowels in Southern Italy: an analysis at the interface between phonology, morphology, and the lexicon

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The Southern Italian dialect of Piedimonte Matese (Campania) shows raising of stressed mid vowels, also known as *metaphony* (cf. Maiden, 1991; among others). In this dialect, while nouns and adjectives show metaphony apparently triggered by word-final high vowels (/u/ and /i/; see *Table 1*), this does not hold for finite verbs (see *Table 2*), where metaphony does not occur in 1SG, although word-final /u/ should trigger it (cf. Gaglia, in preparation).

Table 1 *Nouns & Adjectives*

UR (stressed V)	Word-final /u/ (M.SG) ⇒ <i>metaphony</i>	Word-final /i/ (M.PL) ⇒ <i>metaphony</i>
/ɛ/	/mɛddzu/ → [mɛddzə] ‘half’	/mɛddzi/ → [mɛddzə] ‘half’
/e/	/pɛlu/ → [pi:lə] ‘hair’	/pɛli/ → [pi:lə] ‘hairs’
/ɔ/	/bbɔnu/ → [bbɔ:nə] ‘good’	/bbɔni/ → [bbɔ:nə] ‘good’
/o/	/kɔndu/ → [kundə] ‘bill’	/kɔndi/ → [kundə] ‘bills’

Table 2 *Finite Verbs*

UR (stressed V)	Word-final /u/ (1SG) ⇒ <i>no metaphony</i>	Word-final /i/ (2SG) ⇒ <i>metaphony</i>
/ɛ/	/pɛrdu/ → [pɛrdə] ‘I lose’	/pɛrdi/ → [pɛrdə] ‘you lose’
/e/	/mettu/ → [mettə] ‘I put’	/metti/ → [mittə] ‘you put’
/ɔ/	/mɔvu/ → [mɔ:və] ‘I move’	/mɔvi/ → [mɔ:və] ‘you move’
/o/	/rombu/ → [rombə] ‘I destroy’	/rombi/ → [rumbə] ‘you destroy’

Instead, metaphony regularly raises also the theme vowel of the 2nd conjugation class for 2PL (/e/) although there is no word-final high vowel to trigger the variation.

Example: /e/ → [i]: /vɛrete/ → [vɛ'ri:tə] ‘you see’ (2PL PRES. IND.)

The talk addresses the question how we can account for the asymmetry between verbs, nouns, and adjectives in terms of lexical representations and with respect to the rule format of metaphony. We claim that in Piedimonte Matese, the phenomenon is morphologically triggered for verbs and that the phonological feature [high] of the word-final vowel is not longer decisive at the synchronic level. For nouns and adjectives, instead, metaphony is caused by a morpho-phonological rule. The dependency of the rule format on different lexical categories is caused by diachronic change. Furthermore, metaphony may only be apparently present, being due to the selection of a lexicalized metaphonic stem. Empirical evidence for our claim is drawn from the data collected in Piedimonte Matese. The data will be compared cross-dialectally with data from Cutrofiano (Salento) which also displays metaphony, but where input, output and the environment diverge from Piedimonte Matese. Our hypotheses are tested on data from word-formation.

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Frequency vs. productivity: leveling in the Spanish verbal inflection

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Some of the strongest microclasses within the static morphology of the Spanish verbal inflection are represented by the stress dependent morphonological vowel alternation *e/ie, o/ue* of the type *negár/niégo* ‘to negate/I negate’, *volár/vuélo* ‘to fly/I fly’. Most of the verbs belonging to this alternating class have a high token frequency and as a consequence great stability. However, there is a conduit by which this class has been losing members, namely by generalization of the diphthongal alternant. A classical example is the leveling of the verb *despezár/despiézo* ‘to cut into pieces’ (13th century) into *despiezár* (20th century). González Ollé (1995) analyses this change to the existence of a frequent adjective or noun with a transparent morphosemantic relation to the verb, cf. *pieza* ‘piece’ (11th century). The maintenance of the alternation in the etymologically related, but morphosemantic opaque verb *empezár/empiézo* ‘to begin/I begin’ (12th century) is considered to support this assumption.

There are numerous exceptions, though. An example is the alternating verb *morir/muero* ‘to die/I die’, which does not show any signs of leveling in spite of *muerte* ‘death’ or *muerto* ‘dead’. Another instance is the alternating verb *forzar/fuerzo* ‘to force/I force’, which in Mexican Spanish is being leveled to *forzar/forzo*, despite *fuerza* ‘force’. In view of these contradicting tendencies it is necessary to look for an alternative explanation.

By taking into consideration frequency distribution it will be argued that Dressler’s view (2007) that frequency is subordinated to productivity can successfully be used to explain leveling in Spanish. The principle governing the application or absence of leveling seems to be the productivity content of the respective morphological operation. In this view high token frequency and morphosemantic transparency can contribute to leveling only with the help of productivity. Thus, in order to understand the opposite results of the Spanish alternating verbs it is necessary to take into consideration the amount of productivity of the morpho(n)ological rules involved, i.e. inflectional and derivational. Hereof follows that there is some tension between productivity (= level of competence) and frequency (= performance).

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Syncretism Variation in Gallo-Romance

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The theoretical hypotheses concerning syncretism generally distinguish between two types of inflectional homonymy: take-over and neutralisation (cf. Baerman, Brown & Corbett 2005, Carstairs-McCarthy 1984, Stump 2001). Typological studies like Baerman, Brown & Corbett (2005) and Baerman & Brown (2005) have shown that syncretism is a widespread feature cross-linguistically. W.r.t. verb morphology, the morpho-syntactic categories of PERSON / NUMBER are more inclined to be syncretic than the categories of TENSE, MOOD, and ASPECT. However, within the PERSON / NUMBER category no absolute generalisation can be stated on (im)possible combinations in syncretical realisation, only tendencies may be identified.

This is confirmed by different varieties in the Romance language family. Whereas in Ibero-Romance languages 1 & 3 SG SUBJ, IMPF IND, and COND exhibit syncretism, in Italian all the SG cells in the PRES SUBJ are realised by one single exponent. Gallo-Romance shows a large scale of variation w.r.t. different syncretism patterns and includes those varieties that exhibit the highest degree of syncretism of all Romance languages. A well-known example is the French PERSON / NUMBER syncretism in the present tense: 1, 2, 3 SG, and 3 PL are realised by only one single form.

(1) French /parl/ = {1 SG PRES IND}, {2 SG ...}, {3 SG ...}, {3 PL ...}

The Romance person syncretisms are ambiguous w.r.t. the concepts of take-over and neutralisation due to the interaction of regular sound change the output of which are syncretic forms. In some Gallo-Romance varieties like Lorrain, we can show that in the interaction of syncretism and suppletion (i.e. where sound change is excluded) take-over plays a decisive role in the distribution of the *v-* and the *al-*stem of the suppletive verb “to go” in the PRES IND:

(2) Lorrain /vã/ = {1 & 3 PL}; /ale/ = {2 PL}

Our hypothesis is thus that diachronic spreading of syncretism patterns almost exclusively happens with take-overs.

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Restrictions on Italian verbal evaluative suffixes: The role of aspect and actionality.

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In the grammars of Italian, two seemingly unrelated issues are usually emphasized: Italian is described as a language in which aspect (such as – even if to a lesser extent – actionality) does not play a crucial role in the structure of verb paradigm; and Italian displays a rich and extensive set of nominal and adjectival evaluative suffixes, while, quite surprisingly, verbal evaluative suffixes are very rare and quite unproductive. The hypothesis I formulate in this paper is that there is a correlation between these two issues and that a thorough investigation of the so-called verbal evaluative suffixes provides much information to better understand the role of the categories of aspect / tense and actionality in the verbal systems of the languages concerned. The core of my analysis is based on a sample of approximately 300 Italian verbs containing an evaluative suffix (*-icchiare*, *-acchiare*, *-ucchiare*, *-(er)ellare*, *-ettare*, *-azzare*, etc.) and classified according to their argument structure and to some semantic features which seem to be constant in their meaning (superficiality, habituality, iteration, attenuation, etc.).

In the selection of their domain, verbal evaluative suffixes show a strong preference towards intransitive verbs with aux *avere* (*tossire* ‘to cough’, *ridere* ‘laugh’, etc.) and both transitive and intransitive verbs with aux *essere* (that is, verbs which alternate transitive use with inergative use; *baciare* ‘to kiss’, *scrivere* ‘to write’, etc.). It is not easy to draw a uniform ‘actional’ description of base verbs belonging to these classes, but a clear tendency does emerge: they very often designate events with an atelic (*cantare* ‘to sing’ > *canticchiare* ‘to sing softly, to hum’), dynamic (*trottare* ‘to trot’ > *trotterellare* ‘to scamper, to toddle’) or durative nuance (*dormire* ‘to sleep’ > *dormicchiare* ‘to snooze, to doze’). So, verbal evaluative suffixes exhibit a clear preference towards inergative constructions and, on the semantic ground, towards atelic, durative and dynamic verbs. On the contrary, punctual, stative and telic verbs (*esplodere* ‘to explode’; *credere* ‘to believe’; *morire* ‘to die’) are systematically excluded from their domain of application.

This restriction considerably reduces the domain of verbal evaluative suffixes, but does not explain why even evaluative verbs with an inergative and dynamic / atelic / durative base often show a low and heterogeneous diffusion. The degree of acceptability of evaluative verbs exhibits a considerable range of variation according to the syntactic context, that is if the tense of the verb changes. More specifically, imperfect and the continuous / progressive form of present and past (with *stare* as aux and the gerund of the verb) seem to be the most suitable environment for using a verb containing an evaluative suffix. So, a clear preference of these suffixes towards verbal tenses with an imperfective characterization seems to emerge. The reasons of such a correlation must be looked for in the meaning of evaluative verbs. As I stated before, some semantic features appear to be constant in the meaning of these verbs: superficiality, habituality, iteration, attenuation. Some of these features (iteration and habituality above all) are very close to some values often associated with imperfective aspect (Comrie, B. (1976), *Aspect*, Cambridge, Cambridge University Press, p. 25). Therefore, such a tendency is not surprising: because of the semantic closeness between imperfective aspect and the meaning of the so-called verbal evaluative suffixes, one can easily foresee that their ‘acceptability rate’ is higher with verbal forms usually associated to imperfective aspects than with forms with a perfective nuance.

Data presented so far reveal that the picture about the so-called verbal evaluative morphology is more complex than the one that is normally included in the literature. The constraints on their application and occurrence lead us to question whether these suffixes should really be labelled as ‘evaluative’ or, rather, as markers of aspectual features. Their strict correlation with aspect / tense seems to support the latter hypothesis.

Heads in Compounding: Variation and Patterns

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In this talk we will deal with one of the central aspects in the theory of compounding: headedness. We will show that this notion is far more complex than usually assumed since it can be observed that there is variation not only among languages, but also within the same language and even within the same structural compound pattern. Analysis of the distribution of compound-heads in a Morbo/Comp subsample (Romance, Germanic, East Asian, Slavic, cf. Guevara & Scalise 2007) reveals that this scale of preference is cross-linguistically valid: **Right** > **No head** > **Left** > **Both** (cf. Table 1). As it can be seen, each

group of languages shows a greater or lesser degree of variation with respect to the position of the head. In this talk we will concentrate on the group that exhibits the greatest variation in head position, the Romance languages, and

Headedness	Mean %	ROM%	GER%	E.ASIA %	SLA%
Right	55.89	36.76	80.95	50.85	55.00
No Head	22.05	38.24	12.93	22.03	15.00
Left	9.72	18.38	2.72	8.47	9.29
Two Heads	6.26	5.15	2.04	13.56	4.29

Table 1 Headedness and compound types

on the salient features of headedness, distributed across the most productive structural patterns in this group.

The notion of head has been assumed in the literature with no further specifications. In addition it is usually assumed that each language has a clear preference for one of the two available positions (either to the right or to the left). However, Scalise & Guevara (2006) claim that a more explicit distinction must be made between *formal head* and *semantic head* in compounding and that the two notions can coincide or not. These two aspects of heads have already been discussed in the literature (e.g. Booij 2005, Rainer & Varela 1992, Bauer 1990), although not systematically. We will offer principled criteria to distinguish between the two and show how they can work together. Furthermore, the properties ascribed to heads in compounding reveals crucial in order to characterize the properties of endocentricity versus exocentricity.

Notwithstanding the fact that most compounds are head-driven, some patterns (e.g. N+N attributive comp.) seem to be analogically modifier-driven (cf. Baroni et al. 2007). This leads us to reconsider the theoretical relevance of the head. In particular, it is necessary to consider not only *formal* and *semantic head*, but also what can be called *analogical head*.

As a consequence, the role of specific patterns in determining the canonical head position for a language is crucial. Recent work on Chinese compounding (Packard 2000, Ceccagno & Basciano 2007) suggests that a language may exhibit a different canonical position depending on the structural type under consideration; the same rationale will be shown to apply also to Romance languages. The most important implication of such an analysis is that the position of heads in compounding cannot be considered a linguistic parameter that can be set once and for all.

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The importance of paradigmatic productivity

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Traditionally morphological theory does not deal with phenomena such as blending and clipping. Sometimes these processes are considered to belong to the special field of marginal morphology, but even in this way one cannot account for the overwhelming productivity of paradigms of special kinds of blends.

The examples under (1) may be described as a recurrent process of blending.

(1) infotainment	meditainment	ecotainment	filotainment
edutainment	spiritainment	militainment	
docutainment	relitainment	charitainment	

The derivation would follow the structural description of blend formation as in (2):

- (2) information + entertainment
 A B C D → AD = info + tainment,
 where the resulting form follows the metrical and stress pattern of the original (= *entertainment*)

For the description of the forms under (1) this would mean the rule should be repeated each time, which is ten times here. This is not only not a very efficient way of deriving the forms under (1), but it also does not explain the analogical productivity of this paradigm.

Furthermore rule (2) does not explain why the parts A and D are no words or morphemes.

For the explanation of the first part A one has to include a process of clipping in the derivation. To explain part D a process of reanalysis has to be included as well.

The list under (1) is not the only example of a productive paradigmatic process of word formation. The process is not all marginal.

- (3) commercial (4) advertising (5) broadcast (6) magazine (7) suicide
 infomercial webvertising podcast bookazine homicide
 documercial shockvertising vodvast e-zine familicide
 trashvertising godcast funzine
 petvertising fanzine

In all these examples repeated processes of blend formation may have operated, although in a few of the examples there is no part B. Such an explanation sounds laborious and does not account for the productivity of the paradigms.

In this paper an approach will be defended in which reanalysis of *entertainment* into *enter* and *-tainment* is crucial, together with a process of clipping of *information* into *info*. At the moment one accepts this and similar processes of reanalysis there is a far more satisfactory explanation of the productive paradigm under (1), which is a process of nominal compounding of two clippings, for instance *info* or *edu* and of the semi free form *-tainment*.

In this way there is nothing marginal in the process of paradigmatic productivity, it is simply a matter of reanalysis and analogical word formation.

This process accounts for a great deal of recent neologistic word formation and for changes in morphological processes.

Morphological development in Dutch-speaking cochlear implanted deaf children

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1. Background: cochlear implantation

Today, infants who are born profoundly deaf can access spoken language by means of a cochlear implant (CI), an electronic device that partially replaces the cochlear function via electrical stimulation of the auditory nerve. Implantation in very young children (< 12 months) can be performed safely, providing early access to auditory stimuli.

2. Morphological development in CI-children: Question and State-of-the-Art

In hearing children, grammatical development is surprisingly uniform, both with respect to the time-window in which it takes place¹⁻² as with respect to the fixed order of acquisition of different morphemes³⁻⁴. As the development of oral grammar is at least partially determined by exposure⁵⁻⁶, the question raised here is whether in CI-children, the course of morphological development will be fundamentally different from their hearing peers in both aspects.

Early exposure to language (CI < 12months) is likely to yield overall age-appropriate morphological development⁷. Yet, at the same time, even CI-children who have been implanted during the first year of life may deviate from the usual morphological development patterns⁸.

3. Materials and Method

Our study includes 94 Dutch-speaking hearing-impaired children between 4 and 7 years (44 children with classical hearing aids (HA) and 50 with a CI). A 50 utterances sample of spontaneous speech was analysed according to a standardized test for morpho-syntactic development, based on 240 typically developing Dutch-speaking children⁹. The test addresses 23 morpho-syntactic variables of which we will analyse those that measure nominal and verbal free morpheme omission, as well as agreement errors.

4. Results

The results reveal that both CI-children and HA-children produce utterances that are shorter in length compared to typically developing hearing peers (MLU and MLU5 scores). Comparing the HA- and CI-children, the first show a significant delay (- 2sd) on variables measuring morphological development (omissions and errors in the use of AUX, NPs, DETs, AGR, PastT, PastPart), while the latter perform within the normal range (0 to -1sd). For the CI-population, a positive correlation is found between the children's age at implantation and the morphological variables assessed in this study ($r=0.71$ and 0.69 at the 4 and 6 years' evaluation respectively).

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Inflectional vs. Derivational Morphology in the History of English

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In 1400 years of recorded written history English underwent a typological shift from a predominantly synthetic towards a predominantly analytic language (e.g. Siemund 2004). Evidence for this shift is usually drawn from inflectional morphology, whereas the development of English derivational morphology is usually not included in descriptions of this typological shift of English.

Based on a new quantitative survey of Old and Middle English derivational morphology, we will show that there was a conspicuous trend towards the loss of morphological exponents and the generation of more analytic structures in this domain, too. However, the change is felt less radical here, partly because of the absorption of French- and Latin-based affixes during the Middle English and Early Middle English periods.

In our contribution we will discuss in what way the development of English inflectional morphology differs from that of derivational morphology and present possible reasons for the partly similar, partly divergent development of both morphological components. It stands to reason whether the data support the theory of split-morphology (e.g. Perlmutter 1988) or rather suggest an inflection-derivation continuum (Bybee 1985). Also, one of the puzzles to solve is why language contact apparently could renew derivational morphology, but failed to do so in the area of inflection.

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The development of inflectional patterns in Slavic “contact-subvarieties”: The case of Byelorussian “Trasjanka”

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Bi- or even multilingualism has almost always been a characteristic feature of the territory of present-day Belarus. At least for the last decades especially Byelorussian and Russian have been in “close” coexistence. Today both of them are “official” languages, but Russian still dominates politically and socially, as it has done for more than a century. In everyday communication, though, a large percentage of the Byelorussian population – probably as much as a few million speakers – makes use of a mixed speech variety that displays both Byelorussian and Russian elements and constructions. This variety is commonly called “Trasjanka” (which etymologically refers back to a kind of straw-diluted hay, in other words to bad quality fodder). The contact-linguistic status of Trasjanka is controversial: permanent code-switching, new “mixed” code, extreme borrowing, insufficiently learned Russian (interlanguage), ... Although Byelorussian and Russian are genetically closely related to each other and show a very high degree of structural analogy, inflectional morphology still differs in several paradigmatic positions. Thus inflectional markers and forms of Byelorussian and Russian origin “compete for usage” in Trasjanka discourse.

In my presentation I shall present and discuss several instances of preferences for Byelorussian or Russian or even new, Trasjanka-specific, often “hybrid” forms and markers on the basis of an analysis of their occurrence in a corpus of spoken Trasjanka texts. Possibilities of interpretation of these preferences and obviously emerging new inflectional patterns in Trasjanka will be discussed. The paradigm of demonstrative pronouns of proximal deixis may serve as an example for a new inflectional pattern of clearly hybrid character. Judging on the basis of a pilot study, which is based on 25,000 running words of spoken Trasjanka dialogues, the most frequent inflectional forms of this demonstrative pronoun lack the prothetic consonant *h*, which is characteristic for Byelorussian, but absent in Russian. On the other hand, allomorphy with the alternation of the final consonant /t/ ~ /t'/, characteristic in Russian, but absent in Byelorussian is clearly avoided, cf. the form of the nom. sg. fem. in Br. *hètaja* and Russ. *éta* on the one hand and on the other hand the plural paradigm (nom., gen., dat., ...) in Br. *hètyja*, *hètyx*, *hètym*, ... with Russ. *éti*, *étix*, *étim*, ... and the variant of the Trasjanka from the pilot analysis *ètyja*, *ètyx*, *ètym*, ... Apart from these two rather general tendencies, there are instances of local Byelorussian or Russian “victories”. In nom. pl. as can be seen from the above examples the “long”, Br. ending *-yja* tends to be preserved, and not the Russ. *-i*. In other cases as in the nom. sg. of fem. and neut. gender it is the short, typically Russ. endings (and not the long Byelorussian ones) that tend to be used in Trasjanka conversation: *éta* and *éto* (pronounced identically with a final “a-sound”) rather than *ètaja* and *èta[j]e* (again without the Br. initial *h*). The stability of the long ending *-yja* nom. pl. in Trasjanka is obviously motivated by a tendency to keep the nom. pl. formally distinct from the “Byelorussian like” nom. sg. *èty* (cf. Br. *hèty*), which is clearly preferred over Russ. *étot*. (The latter had developed by a reduplication process when *t_h+t_h* yielded *tot*, later prefixed by the original particle *é*.) In the loc. sg. on the other hand the Russ. form *étom* is dominant over *ètym* (cf. Br. *hètym*) giving preference for the Russ. differentiation pattern loc. sg. ≠ instr. sg. over the Br. one loc. sg. = instr. sg. Special emphasis in the explanations to be proposed will be given to the aspect of inflectional regularity and irregularity in the source languages and in Trasjanka itself.

Hentschel, G. & S. Tesch 2006: „Trasjanka“: eine Fallstudie zur Sprachmischung in Weißrussland. In: D. Stern & Chr. Voss (eds.), *Marginal linguistic identities. Studies in Slavic contact and borderland varieties*. Wiesbaden, 213-243

Morphological change as virtually no change at all: defaults and Latin deponents

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There appear to be cases where the set of morphosyntactic features furnishing terminal nodes in the syntax may be realised by the morphological form not normally associated with that feature set at spell out. Latin deponent verbs present a case in point: *active* morphosyntax required by a particular syntactic construction actually gets the form normally associated with *passive* morphosyntax, as seen in (1) for *hortor* ‘exhort’.

- (1) me=que **hort-antur**
me.ACC=and exhort-3PL.PRESENT.PASSIVE
‘...and they exhort (= ACTIVE) me...’ (Cicero)

Cases of morphological mismatch play an interesting role in language change in that the nature of the change can be seen as an undoing of the mismatch, or a realigning of the syntax with the morphology, as in (2) from Late Latin.

- (2) Episcop-us **horta-batur** a su-is
bishop- NOM.SG exhort-3.SG.PAST.PASSIVE by pronoun- PL.ABL
‘The bishop was exhorted (= PASSIVE) by his men’ (Henry II Chronicle, 1184 AD)

We offer a formal account of morphological change of this type by first separating a lexeme’s syntactic paradigm from its morphological paradigm, and second providing a default relation between the two paradigms that can be overridden, in the spirit of Stump’s theory of Paradigm Linkage (Stump 2006). A lexeme’s *morphological* paradigm is expressed as inheritance from a network of morphological facts. A lexeme’s *syntactic* paradigm is defined as the inheritance of these forms by default for syntax. In the case of *hortor* the syntactic paradigm is distinct from the morphological paradigm because the default is overridden; the relation is changed from syntactic ACTIVE = morphological ACTIVE to syntactic ACTIVE = morphological PASSIVE. Thus the irregularity of *hortor* is expressed as the particular way its syntactic requirement accesses the morphological system. But as a morphological object, *hortor* is perfectly regular, behaving like any other first conjugation verb. The morphological change from (1) to (2) is simply a matter of resetting to default the link between syntactic paradigm and morphological paradigm, allowing the inheritance of passive forms for passive syntax; as well as active forms for active syntax, shown in (3).

- (3) Petr-us horta-bat e-os de cruc-e
Peter- NOM.SG exhort-3.SG.PAST.ACTIVE pronoun- PL.ACC from cross- SG.ABL
‘Peter exhorted them (speaking) down from the cross’ (Gregorian chant, C10 AD).

Language change is usually associated with an innovation: what we didn’t have at one stage we have at another. Illustrating with a range of Latin deponent verbs, we will formalise in Network Morphology (Hippisley 2007) a morphological change completing lacking in innovation. There are no new forms, not even a new ‘exaptive’ way of using them. ‘Change’ here is expressed as resetting the syntactic link to access forms already there virtually, thereby transforming them and their use from virtual to real.

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Variation in the morphology of motion verbs: The case of Romance languages

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In the change from Latin to Romance languages (RLs) verbal prefixes expressing direction lost their productivity and underwent phonotactic and semantic bleaching; moreover, only a small part of the large number of Latin prefixed verbs were inherited by RLs (cf. Stolova in press). The crisis of Latin prefixal system forced RLs to deeply restructure the lexicalization of motion events.

This talk aims at illustrating the consequences of this change in verbal morphology by considering the (lexical and phrasal) strategies RLs have developed to lexicalize motion events.

Recent studies have shown a higher variety of motion patterns than the one expected according to Talmy's classification of RLs as verb-framed languages (a significant typological shift with respect to Latin, which is classified as a satellite-framed language).

The variation in the morphology of Romance motion verbs concerns both diachrony and synchrony. For instance, French has passed through a verb-particle phase in the mediaeval period (Dufresne, Dupuis & Tremblay, 2003); Italian has recently accepted such a construction in the standard language (which was confined to dialectal and sub-standard varieties for a long time), which construction has become the main way to productively express verbal motion events: e.g. *venire giù*, lit. come down 'to come down', *portare via* lit. take away 'to take away', *mettere sotto* lit. put under 'to run over' (cf. Iacobini & Masini 2007).

As far as synchrony is concerned, usage-based studies have shown that encoding of path in the verb stem is only one of the possible strategies adopted by Spanish and French (Kopecka 2006). The satellite-framed pattern is pervasive for the expression of path of motion in Rhaeto-Romance dialects (Kramer 1987) and in other 'minor' Romance varieties (Cini in press).

The theoretical approach of this talk refers to Construction Morphology, since it aims at giving an integrated account of the variety of morphological, lexical, and phrasal ways of expressing motion in RLs. This talk also argues against the classification of language families in typological global types.

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Natural Morphology approach to suppletion in Late Old English and Early Middle English

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The paper investigates the problem of the retention and the replacement of suppletive forms found in Late Old English and Early Middle English texts. In particular, the issue of frequency and its role in the retention or the replacement of individual suppletive forms will be addressed. It is often claimed that forms which appear to be more frequently used in language are frequently retained. This is the view expressed, for instance, by Dressler (1985). The approximate frequency of particular forms is calculated using the data from the Dictionary of Old English project. In the present study individual Late Old English and Early Middle English texts are examined in full. Using the Corpus Presenter software Early Middle English texts such as the Peterborough Chronicle, the Ormulum, the Ancrene Riwe and others are investigated. The data found in these texts provides sufficient information on the process of the retention and the replacement of suppletive forms. Another problem investigated in the study is the chronology and the dialectal distribution of particular suppletive forms in Middle English. Thus, the study provides the chronology of the use and the dialectal distribution of suppletive forms of verbs, adjectives and numerals. For example, with regard to the suppletive comparative adjectival forms the *Oxford English Dictionary* states that the s-forms (*selra~selest*) survived until the thirteenth century. However, there is little information in the previous literature about the details of the replacement of the s-forms by the rival b-forms (*betera~best*). It will be argued that the replacement of the suppletive forms proceeds in accordance with the universal claims of Natural Morphology.

Variation within Voice Patterns: Minding the Gaps

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While voice syncretisms (i.e. cases in which the verb appearing in different syntactic frames bears the same voice morphology, as in (1) for the passive and inchoative frames of the Albanian counterpart of ‘break’) have been the subject of substantial research, little attention has been paid to voice gaps. These are cases in which verbs within one and the same language differ in the voice morphology they may bear within the same syntactic frame, as in (2a,b) for ‘crack’, both of which may have an inchoative reading irrespective of the fact that in (2a) the verb bears non-active while in (2b) it bears active voice. The basic observation is that within and across languages with morphological voice distinctions (here specifically: Albanian, Greek, Latin) while passive verbs *cannot* appear in active voice, or transitives and unergatives in non-active/passive (depending on the language) voice, inchoatives may be: (i) exclusively non-active for a given verb; (ii) exclusively active for a given verb; (iii) (optionally) either non-active or active for a given verb; (iv) mixed paradigms for different tenses for a given verb and/or for different dialects. That is, while passives and transitives/unergatives are morphologically robustly marked cross-linguistically (cf. also Haspelmath 1990), voice gaps (and thus voice-related variation) across languages arise *only* with inchoatives. This situation challenges the most basic and widely adopted claim on non-active voice, namely that it relates to just a [-external argument] feature in the syntax: since the absence of an external argument does not entail non-active voice (as revealed by patterns of types (ii) to (iv)), the correlation between non-active voice and lack of an external argument is at best an imperfect one. What then is the feature that non-active voice relates to and that distinguishes it from the active voice? The central claim that I put forward in this paper is that the (syntactically relevant) feature realized by non-active morphology is (or is being re-analysed as) a fundamentally aspectual one. Specifically, I will show that a variety of facts across languages converges on the view that non-active/passive voice relates to a syntactic (and therefore also semantic) prime that identifies the ontological event type of the predicate as [+sustained activity] or [+open-endedness] in the presence of a [-external argument] feature.

Examples:

(1) *Vazoja* **(u)* *thye.* (Albanian)

vase.the_{NOM} NACT broke.AOR.3S

(i) ‘The vase was broken.’

(ii) ‘The vase broke.’

(2) a. *Dritarja* *u* *kris.* vs. b. *Dritarja krisi.* (Albanian)

window_{NOM} NACT crack.AOR.3S window crack.ACT.AOR.3S

(i) ‘The window cracked.’

(i) ‘The window cracked.’

(ii) ‘The window was cracked.’

(ii) *‘The window was cracked.’

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Deponents across Albanian, Greek and Latin: Variation and Commonality

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1. Background: A major challenge for detransitivization approaches to voice alternations has been the existence of morphologically *non-active* or *passive* verbs (depending on the language) that have no *active* voice counterparts. (We label these verbs ‘deponent’, drawing however attention to the fact that this term is overcharged since traditional grammars of Latin/Greek also list ‘transitive’/‘active’ deponents that may combine with accusative objects – a situation which in Albanian, on the other hand, is impossible, non-active voice and transitivity/accusative case being incompatible.) At first sight then the existence of deponent verbs lends support to transitivity approaches to voice alternations, as the idea that voice morphology is completely otiose (Campos 1987) is problematic. However, transitivity approaches are also subject to criticism on account of other considerations (Pylkkänen 2002).

2. Novel facts: Crucial evidence against the superfluity of non-active voice marking in deponents is the fact that we found *all* deponent verbs in Albanian (a language with two distinct conjugational paradigms, active/non-active) to be either denominal or de-adjectival, replicating in this way results in Xu, Aronoff & Anshen (2007) on Latin. Our second finding is that deponents at least in Albanian/Greek *cannot* combine with either agentive, or causative PPs (i.e. *by-* or *from-*phrases). This renders untenable the claim in Alexiadou et al. (2006, 2007) that non-active voice is solely of two varieties, viz. VOICE [+Ag] and VOICE [+Caus].

3. Proposal: The central claim that we put forward in this paper is that irrespective of differences in their syntactic behaviour, deponent verbs across Albanian/Greek/Latin are uniformly backformations (from nouns or adjectives). This raises the following questions: (i) If a verb is deadjectival or denominal, can it therefore be concluded that it is deponent (i.e. is the correlation between deponency and denominalization/de-adjectivalization two-way)? (ii) If not, is there a difference in the productivity/transparency of the nominal/adjectival source (Kiparsky 1997; Pinker 1999 on irregular inflection)? (iii) Is backformation a synchronically relevant phenomenon afterall (contra Aronoff 1976)?

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English compounds in the light of idiomaticity

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Difficulties we encounter when trying to set forth a proper definition of *compounds* make me propose an alternative view, namely treat the category as idiomatic expressions. As a matter of fact, compounds as we understand them share with idiomatic expressions quite a few features and properties, including the ways of rise, existence proper, and interpretation. Most arguments, based on a fairly extensive research, are introduced in my chapter as a part of *A Handbook on Compounding*, which is being prepared by R. Lieber & P. Štekauer (eds.). For reasons of space in the book, some ideas are worth pointing out again, and some more should be added, too.

Although perhaps taken for granted, we tend to use the term ‘compound’ to name an expression coming into existence through combining two or more simplex words (constituents), finally representing a unit syntactically and meaningfully invariable in the current context. The role of context (in its broader sense, i.e. both verbal context and context of situation) should be stressed here: namely, I believe that compounds cannot be treated strictly as isolated phenomena (of grammar or lexicon); their status will be understood more readily if they are viewed as parts of concrete, contextually defined utterances. Let me point out too that the aforementioned tentative definition of compound also implies the idea of process: on the one hand, it makes it possible to look upon syntactic strings (free combinations, e.g. *wooden desk*) as potential compound candidates; and on the other hand, it sets grounds for the meaning processing, pragmatically viewed as a (very often creative) coinage of one specific meaning on the speaker’s part (e.g. *blue blood*) and the choice of (usually more than one) possible interpretation(s) on the receiver’s part (e.g., *red carpet*). While on the speaker’s part we deal with the phenomenon of ‘compositionality’, on the receiver’s part we zero in on that of ‘predictability’. (The two phenomena are certainly interdependent; they are not in a truly proportionate relation, though.) Put in psycholinguistic terms, the new, namely the total meaning is arrived at by computation of meanings of respective constituents, either directly, or through figures of various types (metonymies, simple metaphors, underlying metaphors, etc., e.g., *the White House*; *stone-fly*, *green thumb*; *greybeard*). What definitely sets off compounds from free combinations is the complex phenomenon of compositionality: like idiomatic expressions, compounds are non-compositional, namely, restrained in (1) their lexical and syntactic variability (flexibility, modification), e.g., **the snow-White House*; **living cozy room*, (2) and / or literalness, being interpreted ‘literally’ (e.g., *washing machine*), ‘non-literally’ (e.g., *white lie*), and ‘figuratively’ (e.g., *red tape*). It only remains to add one important thing: respecting the dynamic character of language, we must look upon compositionality as on a scalar, gradient phenomenon. This shows itself in two levels, traveling on two axes, as it were. On the vertical axis, say, the ‘synchronic’ one, we can consider sometimes fine degrees of literalness (specifically where free combinations border on idiomatic expressions): suffice it to focus on, e.g., *footbridge* – *life boat* – *country house*. On the horizontal axis, in fact representing a diachronic change, we can illustrate the potentiality of the rise of compound from a free combination: e.g., *dark+room*; *dark+horse* (the two being simultaneously on different points of the vertical axis). Frequency of use, familiarity of the novel expression – in fact what we name ‘conventionality’ speaks in favor of the hypothesis that any free combination makes a compound candidate. (Although the following example may sound too progressive a vision, we cannot deny that *blue hotel* used in an appropriate context could be interpreted figuratively, like already is *blue joke*.)

Similarity and productivity in inflectional morphology: A comparison of minimal generalization and memory-based learning.

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I will compare memory-based learning (MBL, Daelemans and Van den Bosch, 2005) to minimal generalization learning (MGL, Albright and Hayes, 2003), a rule-based model, in the domain of English past tense inflection. In both approaches novel inflected forms -regular or irregular- are produced by a single similarity-based system, as opposed to dual route theories in which the production of regular forms is considered a symbolic process.

I will first present simulations showing that MBL achieves good fits to data from experiments (Prasada and Pinker, 1993) in which participants rated the likeliness of regular ($r=.69$) and irregular ($r=.55$) past tense forms of nonce verbs, whereas dual-route theories predict that similarity-based models should only fit well to ratings of irregular past tense forms. MBL is then compared to MGL, which Albright and Hayes (2003) claim is superior in identifying islands of reliability for inflectional patterns. This claim, based on a comparison of MGL to the generalized context model (a memory-based model), will be disputed on methodological, empirical, and theoretical grounds. First, I will show that an apparent advantage of MGL over the generalized context model is an artefact of an oversight in scaling model results to experimental data. Second, I will present results from simulation studies showing that MBL achieves better fits on the Albright and Hayes data than the rule-based model. Finally, I will show that MGL can be implemented as a memory-based model giving identical results. On the basis of these findings, I will argue that productivity in inflectional morphology is data-driven and that MBL and MGL are implementational variants of this position.

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Areal-typological aspects of word-formation. The case of *aktionsart*-formation in German, Slavic, Yiddish, Hungarian and Romani

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The notion of *aktionsart* is defined as the modification of verb meaning by morphological means. *Aktionsarten* are typical of Russian, Yiddish, Hungarian but not of English, French, Romanian and only to a lesser degree of German.

It will be claimed that *aktionsarten* are an areal phenomenon largely brought about by contact with Slavic. Most *aktionsarten* are expressed by prefixation and only a few by suffixation. We will discuss *aktionsart*-formation by means of prefixation only.

The most spectacular influence of Slavic can be observed in Yiddish where some of the Middle High German prefixes/particles have been reinterpreted to express Slavic-like *aktionsarten*. At the same time *aktionsart*-formation has become (more) productive and some new *aktionsarten* have been created.

Hungarians came into contact with an important Slavic population when they occupied the Carpathian Basin in the 9th century. More and more particle verbs came into existence to express aspectual functions. At the same time Slavic-type *aktionsarten*, too, appeared. Slavic influence may have played an important role at the beginning of this process. The development of particle verbs with aspectual and *aktionsart* meanings is due to the breaking down of the old Hungarian tense system.

Until the end of the 19th century Romani retained the old Indoeuropean tense system and had no prefixed or particle verbs. The first complex verbs were registered during the first half of the last century. Romani verbs have acquired different morphological structure in Hungarian, German and Slavic context. Contact with Hungarian led first to the development of V + Particle constructions, in which Romani adverbs were used as particles. The only exception was the opaque particle *meg*, which was simply borrowed. Similar things can be observed in the case of contact with German. Finally, Slavic verbal prefixes were felt to be semantically opaque and were prefixed to Romani verb stems. Aspect as well as *aktionsarten* are expressed by means of complex verbs.

Against a derivational analysis for pluractional verbs in French and Italian

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Pluractional verbal derivation is a reputedly “exotic” phenomenon. Examples are reported from such languages as Paiute, Niuean, Chechen, etc. (see Yu 2003 ; Haji-Abdolhosseini et al. 2005; Houser et al. 2006). Pluractionality has been studied for English where simple verbs with a pluractional meaning do exist, e.g. *nibble* vs. *bite* (see Cusic 1981; Lasersohn 1995).

Such verbs are also well represented in French and Italian: cf. *mordiller* /*mordicchiare* ‘nibble’ vs. *mordre* / *mordere* ‘bite’. On the semantic side, they show diagnostic features of pluractionality: successive repetitions at phase level within the confines of one event, and reduction of the amplitude of the process. Morphologically, they seem to be deverbal derived verbs, inviting analyses such as /*mord-ij-e*/ and /*mord-ikkj-are*/, with /*ij*/ and /*ikkj*/ pluractional derivational suffixes – called ‘diminutive’ or ‘pejorative’ in traditional grammars.

We will argue against such an analysis. Our main reasons for not regarding these verbs as deverbal derivations are the following. First, many of them have no simple counterparts: cf. French *grappiller* ‘to glean’ (no **grapper*), Italian *balbettare* ‘to stammer’ (no **balbare*). Secondly, the apparent suffixes are quite numerous: 9 in French and 14 in Italian (at least), with no rationale for selection on individual verbs – although native speakers have clear-cut judgments about possible forms (e.g., *mâchouiller* ‘to munch’ but **grapouiller*, *fumacchiare* ‘to smoke desultorily’, but **balbacchiare*). Thirdly, leaving these verbs aside, deverbal derivation in Romance is only through prefixes (cf. *faire* / *défaire*, *mettre* / *smettere*). Fourthly, all these verbs belong to the 1st conjugation class, whereas paired simplexes (when they exist) appear in all classes. Conjugation class in French and Italian is a property of the simple stem (prefixation doesn’t modify the class), which suggest that our endings are included in it.

We will therefore assume that /*mordij*/ and /*mordikkj*/ are underived stems, which means that /*ij*/, /*ikkj*/, etc. are not morphemes. Yet, they do fulfil a special function for interpretation, namely signalling that the internal structure of the event is characterised by a multiplication of phases with compensatory partial affectedness of the patient argument, which is the specificity of these verbs in our opinion. The contradiction can be resolved by viewing them as submorphemes, i.e. phonological strings without a meaning, but inducing meaning effects through their phonetic substance (cf. /*sl*/ at the beginning of English *slip*, *slide*, *slither*, *slobber*, etc.). Pluractionality is then a feature of the whole form, which is flagged, although not specifically expressed, by the phonetic form of the ending. This accords well with the semantic nature of pluractionality, not an added specification, but an inherent modification of the process denoted by the simple verb the pluractional verb shares its root with. It also accounts for why there need not be a simple verb.

Finally, our analysis enters the Word-and-Paradigm framework quite naturally. Only this framework, we think, is able to give submorphemes the not so marginal place they deserve in morphological theory.

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Psycholinguistic Evidence for DATR-based Descriptions of Inflection

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The goal of our work is to develop an empirically based theory of German inflection, using both psycholinguistic and computational-linguistic tools. For our representations we employ DATR (cf. Evans & Gazdar 1996), a formal language for computational descriptions of inflectional morphology, where nodes typically describe lexemes and inflectional types, while inheritance relations capture the features common to distinct types. With its nonmonotonicity DATR can represent default relations between regular and irregular inflectional classes regarded as abstract linguistic entities and can express hierarchical markedness relations of forms within the paradigms of individual classes.

But since DATR-based descriptions so far have not made empirical claims going beyond descriptive correctness of the inflected word forms predicted by the analysis, we want to capture generalizations about language acquisition or change which might shed light on the mental organization of inflectional systems.

We express the findings of psycholinguistic studies formally so that we can establish whether overgeneralizations and patterns of variation are systematic and constitute a partial ordering of the inflectional types which can appropriately and adequately be expressed in the nonmonotonic inheritance hierarchies of DATR. If hierarchical ordering of this kind can be supported empirically, it would speak against a simple dichotomy of regular versus irregular inflection; if not, and if irregularities prove to be unstructured, then DATR would appear to be overly expressive from the viewpoint of empirical linguistics.

Furthermore, we intend to determine whether the overgeneralizations found for a lexeme always constitute a systematic shift to another existing inflectional type (e.g. *singen* –* *singte* – **gesingt* analogically to regular *lachen*), whether new paradigm types can arise (e.g. *singen* – *sang* – **gesingt*), or whether paradigms are not yet established in the speech of young children. The latter question is directly relevant for the formal issue as to whether DATR-based morphological descriptions should allow virtual nodes for inflectional types not corresponding to concrete lexemes.

We have chosen DATR as our representation language because it is more expressive than other frame-based formalisms such as typed feature structures, which is due in particular to DATR's rich variety of means for stating nonmonotonic inheritance relations.

In sum, we aim to establish an integrative approach in which we conduct psycholinguistic and CL studies of clearly defined phenomena and relate the results to each other.

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Formal variation in Modern Hebrew Lexeme Formation: the case of /ut/ ending nouns.

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1. Object. Nouns ending with /ut/ (thereafter *Nut*) in Modern Hebrew (MH) are mostly described as abstract nouns. Usually, in new coined *Nut* /ut/ is analyzed as a suffix, probably by analogy with European processes of lexeme formation (LF) [1]. Conversely, attested *Nut* are known to be formed either by /ut/ suffixation, or by means of a template (e.g. $C_1C_2iC_3ut$). My aim is to propose an analysis of *Nut* LF in the lexematic morphology (LM) framework [2-4], providing a semantic enlightenment to the current morphophonological point of view [5].

2. Description of MH LF: root-pattern vs word-based. LF in Semitic languages is usually presented by means of consonantal *roots*, the lexical content, and vocalic *patterns* of derivation (e.g. $C_1aC_2aC_3$ (G-D-L) = gadal ‘he raised’). Recently, some linguists (e.g. [6]) started to assume a word-based approach to describe morphophonology of MH, within the *Stem Modification Theory* which eliminates the notion of the root from the grammar of MH: indeed, for each complex lexeme a lexematic base can be found.

Rule-based approaches in LM offer a multidimensional description of lexemes (phonetic, syntactic and semantic). A single semantic relation between lexemes related by a rule can correspond to several formal realizations:

Rule	Base lexeme		Constructed lexeme: agentive N
LFR-an	KALKAL _N ‘economy’	→ _{-an}	KALKALAN _N ‘economist’
LFR-aj	‘ITON _N ‘newspaper’	→ _{-aj}	‘ITONAJ _N ‘journalist’
LFRconv	SAPAR _V ‘to do sb’s hair’	→ _{conversion}	SAPAR _N ‘hairdresser’

3. Nut

3.1. Schwarzwald & Bolozky (S&B) Analysis. S&B describe the *Nut* formation from a morphophonological point of view only. S&B assume a double formation system both ‘linear’ (i.e. suffixed) and ‘discontinuous’ (i.e. template) according to whether the base and the stem of the *Nut* are identical or not (e.g. BALŠAN ‘linguist’ → BALŠANUT ‘linguistics’ vs TAMIM ‘innocent’ → TMIMUT ‘innocence’).

3.2. Lexematic Approach. My analysis of the *Nut* corpus from [5] leads to the observation that *Nut* list members belong to six main semantic classes (e.g. ideologies, properties), and that *Nut* bases belong to seven other classes (e.g. agentive N, adjectives). I wish to show that a unique LF rule builds all *Nut* corpus members. This rule only semantic contribution is that of a “nominalizator”, through the two formal realizations proposed in [5]. The *Nut* meaning entirely depends on the base’s semantic properties: e.g. the neologism SANDLARUT (shoemaking) is recognized as a professional activity Noun, because of the agentive property of its base SANDLAR (shoemaker).

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Productivity, rule variation, and rule generalization

(The case of the Hungarian derivational affix *-(s)di*)

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While in Kiefer–Ladányi (2000) *-(s)di* is not included into the list of productive derivational suffixes, Keszler (2000) states that there are two main and productive ways of forming new nouns by it: denominal and deverbal suffixation. Since Keszler’s statements have not been proved by either a reasonable amount of data or acceptable arguments, the status of the suffix needs further investigation. One of the goals of the article is to provide the missing data from the Hungarian National Corpus and (on the basis of a qualitative approach to the notion) argue for productivity of *-(s)di*.

Another question discussed in the article is whether we really need two separate derivational rules for *-(s)di*. This question is a complex one, including both the form of the suffix (*-sdi* or *-di*) and the class(es) of the possible base words: a) nouns and present participles or b) adjectives derived from them by suffix *-s*. The author will argue for the *-di* form of the suffix, derived adjectives as base words, and one derivational rule $[[X + -s]_A + -di]$.

This solution is different from the traditional one we find in Keszler (2000). However, forming the two different rules into a single one is not only more economical but also provides a higher degree of generalization.

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The Morpho-phonology of Hebrew Valence Changing

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This talk sheds light on the interaction between valence changing operations and morpho-phonological constraints in Hebrew. I examine the thematic operations that derive decausatives, reflexives and reciprocals, manifested in Hebrew by relations among prosodically distinct configurations, called binyanim (e.g. *niCCaC*, *hitCaCeC*). Following Reinhart & Siloni (2005), I assume these operations apply in the lexicon, while passivization applies in the syntax. I argue that the criteria for choosing a binyan are unique to the morpho-phonology of the lexicon, thereby supporting the existence of two types of morpho-phonology. The derived counterparts (e.g. decausatives) of *hiCCiC* verbs exhibit an intriguing variation. Some are formed in *niCCaC* (e.g. *hirdim* ‘put to sleep’- *nirdam* ‘fall asleep’), while others are formed in *hitCaCeC* (e.g. *hirsim* ‘impress’- *hitrašem* ‘become impressed’). This variation results from two competing morpho-phonological constraints. The selection of *hitCaCeC* is due to **markedness**; *hitCaCeC* is less marked than *niCCaC* as it is prosodically uniformed throughout its inflectional paradigm. The selection of *niCCaC* is due to **faithfulness**; *niCCiC* has a consonant cluster in its past and present forms, which allows preserving the structure of the input of *hiCCaC*. Choosing *niCCaC* yields (partial) uniformity within the derivational paradigm. Moreover, there are four patterns that manifest a tendency to choose *niCCaC*. This stems from faithfulness constraints that block the application of phonological processes.

1. Block deletion: Verbs whose initial stem consonant is *t* or *d* usually escape *hitCaCeC*, since such derivation creates the homorganic /tt/ or /dt/ clusters, which are prohibited in Hebrew. Forming a verb in *niCCaC* blocks the application of deletion, which would be required to amend in homorganic clusters. For example, the decausative counterpart of *hidhim* ‘amaze’ is *nidham* ‘become amazed’ and not **hitdahem* or **hidahem*.

2. Block metathesis: Some verbs with a strident as their initial stem consonants do not have a derived form in *hitCaCeC*, as this would result in metathesis (e.g. *hitsarek* → *histarek* ‘comb oneself’). Again, the selection of *niCCaC* allows avoiding the application of this process (e.g. *hicmid* ‘stick’ → *nicmad*/ **hictamed* ‘stick oneself’).

3. Block prosodic and vocalic alternation: Verbs whose initial stem consonant is a glottal stop have an identical prosodic structure in *hiCCiC* and *niCCaC*. The glottal stop in both binyanim is preceded and followed by /e/. For example, *he’eliv* ‘insult’ and its derived counterpart *ne’elav* (**hit’alev*) ‘become insulted’ share the same number of syllables and two first vowels. *niCCiC* is more faithful to *hiCCiC* than *hitCaCeC*, and hence it is preferred.

4. Block stop-fricative alternation: Verbs whose first consonant stem is /f/, /v/ or /x/ usually have a derived counterpart in *niCCaC* in order to maintain the fricative consonant. As these fricatives tend to surface as stops in post-consonantal position, derivation in *hitCaCeC* may result in an alternation (e.g. *hivhil* ‘frighten’ → *nivhal* / **hitbahel* ‘become frightened’).

The analysis reveals the effect of morpho-phonological constraints on thematic operations. In contrast to the above constraints, the morphology of passivization is predictable and is hardly subject to variation. I argue that this difference results from the component of the grammar where operations apply. As the above constraints are unique to the lexicon, the analysis supports the claim that morphology is an independent component that interacts with the lexicon (Aronoff 1976, Anderson 1977, Scalise 1984, Borer 1991 among others).

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Phonological triggers of the morphological change

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The paper discusses the impact of phonological changes on inflectional morphology. The conflict between the phonological development and the communicative demands of a morphological system of a language is one of the most important factors triggering morphological (therapeutic) changes. Several instances (taken from Slavic data) of phonologically elicited morphological development of this kind are considered, and some general trends governing the process are elucidated.

One of the most interesting cases is the evolution of the category of gender in Slavic languages, viz. the development of the category of animacy /personality. The development was triggered originally by the Nom.-Acc. syncretism of masculine nouns in the late Proto-Slavic, which was due to the phonological development in the word coda. The tendency to reconstruct a morphologically marked opposition between the Agent and the Patient function had provoked a morphological drift. The drift extended in the Slavic area under almost 8 centuries. A noteworthy feature of the drift in question is its cyclic character and the fact that the internal structure of a cycle (i. e. consecutive steps of the development) is repeated at each phase of the process. The resulting typology of the Slavic gender systems mirrors the phases of this development.

Interesting examples of therapeutic morphological changes are also attested in the developments of the conjugational systems of different Slavic dialects. Phonological changes such as denasalisation of nasal vowels, changes of unstressed vowels etc. could have been destructive to the morphological person and number system. Several types of morphological changes preserving the functional structure of the system are at stake here. An attempt at a typology of such changes is presented in the paper.

Stem Frequency and Type Numerosity in visual word recognition of Italian words

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Previous studies on a variety of languages (Dutch, English, Hebrew, Finnish) have shown that the number of word types in a morphological family (Family Size) predicts the word recognition latencies better than the frequency of the morphological family itself. The family size effect is also influenced by the semantic consistency of the base over the family members.

The aim of the present research was to provide a test of the frequency of the inflectional and derivational family in the recognition of verbal forms. We also aimed at exploring possible effects of Family Size in Italian, a language with a smaller average number of family members.

Three experiments based on the lexical decision task were carried out. In Experiment 1 we selected two groups of verbal forms with high vs. low family frequency. The two lists were matched for mean length in number of letters, mean surface frequency, mean summed frequency of the inflected words, mean summed frequency of the derived words, and mean family size. In Experiment 2 we selected two groups of verbal forms with high vs. low inflectional frequency. The two lists were matched for mean length in number of letters, mean surface frequency, mean summed frequency of the derived words, and mean family size. In Experiment 3 we selected verbal forms belonging to three different ranges of family size and with two different degrees of semantic transparency. The different lists were matched for all the frequency parameters.

The Anovas on both latencies and errors did not show any stem frequency effect, neither for inflectional frequency, nor for derivational frequency. The main effect of family size emerged only for items belonging to semantically consistent families.

Our results extend the family size effect to Italian, and provide a distributional function of the effect over families of different size. We hypothesize that this effect could also be crucial to explaining the inconsistency of the stem frequency effects.

The representation of gender information in the Italian mental lexicon

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In Italian phonologically transparent nouns the information about both gender and number is expressed in the inflectional ending.

The main aim of this present study is to investigate the role of the gender information in the processing of a particular kind of transparent nouns, namely those which have four inflected forms (e.g. *nonno* (grandfather), *nonna* (grandmother), which are the singular masculine and the singular feminine form, respectively; *nonni* (grandfathers) and *nonne* (grandmothers), which are the plural masculine and the plural feminine forms, respectively).

There are different psycholinguistic and linguistic theories which are in disagreement about the lexical representation of these nominal forms; some of them claim that the lexical entry is represented by the citation form, namely, the singular masculine. Other theories claim that forms belonging to different grammatical genders are independently represented.

Our hypothesis is that in the mental lexicon these nouns are organized according to the gender information, with independent representations for the masculine and feminine forms.

A free recall task was employed as the experimental paradigm. The analysis of correct recalls and errors show a statistically significant tendency to produce singular instead of plural forms. Moreover, our data show a comparable pattern of correct recalls on masculine and feminine singular forms. The results support the view that these nominal forms have two distinct representations, one for the masculine gender and one for the feminine gender.

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Morphology across registers: a corpus-based approach to English prefixation

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This poster examines the impact of register variation on English prefixation. It reports the results of a corpus-based analysis of English prefixes in three written registers: press editorials, academic prose and fiction. The study is part of an on-going project in corpus-based contrastive morphology.

Increased attention has recently been paid to register variation in studies on word-formation (see e.g. Plag et al. 1999; Grabar et al. 2006). However, these studies have mainly focused on particular derivational suffixes. Prefixes have tended to be underexplored in word-formation studies in general (Lehrer 1995) and in register variation studies in particular. The present study aims to give prefixes the attention they deserve.

This poster is divided into two parts. The first section is devoted to the methodology adopted. Potential prefixed words were extracted automatically with the help of a PERL program and subsequently filtered on the basis of a range of explicit criteria, among others: the nature of the affixed elements, analysability and the nature of the base or root. This filtering process, which is crucial in selecting the relevant data, proved to have a tremendous impact on the results. It was carried out manually, as automatic tools cannot yet readily yield highly reliable results. The second part of the poster presents the results of the corpus-based analysis in three written registers. Three English subcorpora were used: (1) TRILLED (*Trilingual Louvain corpus of Editorials*), a corpus of press editorials; (2) KIAP (*Cultural Identity in Academic Prose*, Fløttum et al. 2006), a trilingual academic corpus of research articles from economics, medicine and linguistics; and (3) the fiction part of the bilingual English-French PLECI (*Poitiers-Louvain Echange de Corpus Informatisés*) corpus. Approximately 100 English prefixes have been investigated in the three registers in terms of their extent of use (as a first rough indicator of their possible productivity).

The study paves the way for a series of follow-up studies in the field of register variation in word-formation.

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Verb prefixation in Polish: A Construction Grammar Approach

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In this paper we provide a semantic account of Polish perfectivizing prefixes based on both Goldberg's (1995) constructional approach to English resultative constructions and Michaelis & Ruppenhofer's (2001) constructional approach to German *be*-prefixation constructions. Our goal is: (i) to show the theoretical and empirical superiority of our account of verbal prefixation in Polish over Spencer & Zaretskaya's (1998) projectionist analysis of verb prefixation in Russian; (ii) to contrast our Construction Grammar approach with Dąbrowska's (1996) Cognitive Grammar account of verbal prefixation in Polish, showing that the former is more appropriate when dealing with the important distinction between 'constructional meaning' and 'verbal meaning'. We argue that S&Z (1998) are wrong when saying that the perfectivizing prefix is sometimes "a simple aspectual marker of telicity": they ignore the important prototype-based effects involved in Slavic verbal prefixation (e.g., cf. Dąbrowska (1996)). Moreover, unlike S&Z (1998), we argue that both the syntactic effects (e.g., the "unselected" direct objects in Polish examples like *Wypisałem długopis* lit. 'out-I-wrote the pen', i.e. *My pen has run out of ink*) and the semantic effects (e.g., the perfective contribution of the prefix) do not come from lexico-conceptual operations on input verbs but are the products of a reconciliation procedure in which the verbal meaning is integrated into the event-structure schema denoted by the prefixal construction. On the other hand, following Pasich (1993) and Dąbrowska (1996), we reject the idea that there is a single abstract meaning associated with perfectivizing prefixes. For example, when dealing with our constructional analysis of the Polish prefix *prze-*, we propose to capture the commonalities among usages of it through an associative network based on a semantic schema where a Figure physically crosses the Ground. We also argue that this schema is the basis for certain metaphorical extensions: following Goldberg (1995), we propose that the relation between the event structure schema and these metaphorical extensions are represented by links denoting metaphorical mappings (cf. Dąbrowska 1996: e.g., TIME IS SPACE (cf. *Przetańczyliśmy całą noc* lit. 'across-we-danced whole night', i.e. *We danced all night*)). Quite importantly, we argue that Dąbrowska's (1996) claim that the prefix makes the verbs transitive by strongly evoking a landmark elaborated by the direct object can be provided with more explanatory value if one assumes Goldberg's (1995) or M & R's (2001) CG framework: indeed, it is not the case that the prefix is added to the verb changing then its valency (as Dąbrowska claims) but rather it is the verb that is *integrated* into a transitive construction headed by the prefix. Accordingly, we argue that Goldberg's (1995) or M&R's (2001) CG framework is more appropriate since it allows us to provide an elegant model of the valence-creation (cf. *Podpiwniczyli dom* lit. 'under-they-cellar-ed the house', i.e. *They put a cellar under the house*) and valence augmentation (*Zakrzyczeli Jana* lit. Behind-they-shouted Jan, i.e. They shouted louder than Jan) functions of the prefix, which are harder to account for in verb-centered models.

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CONSTRAINTS AND ANALOGY IN THE LEXICON

An analysis of adjectives derived with *-ien* and *-iano* in French and Italian

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Our assumption is the fact that a speaker's lexical competence depends on at least two factors: the memorized lexicon, by nature composite and stratified, and word formation rules/patterns. The former is only partially constituted by units constructed by means of the latter, since it contains borrowed words or words constructed at previous stages in the language. We consider that surface forms of words constructed by synchronically productive rules result from the interaction of various constraints. These constraints are principally phonological, but phonology alone cannot always explain the shape of morphologically complex word. Similarity with words belonging to the same morphological family can be as important as the non violation of phonological principles. Our data are taken from a large database of derivatives not recorded in dictionaries, and mostly retrieved via Web searches.

Suffixes derived from Latin *-(i)anus* in French and Italian provide good examples of constraint interaction: first, in both languages, several variants exist for those suffixes. Second, a variant can be chosen in order to satisfy some constraints. Interestingly, the same constraints apply in French and Italian, although in a different order. In both languages the variant displaying glide [j] (*-iano* and *-ien*) is nowadays most common. However new words construed without glide are quite frequent. The choice between allomorphs is conditioned by: (a) phonetic constraints, such as the tendency to avoid some phonemes sequences; (b) the need to maximally preserve the segmental material of the base (c) size constraints (d) pressure from the existing lexicon: e.g. the variant *-ano* can be considered as unproductive nowadays, although it was productive in the past, the Italian lexicon including many words definitely analyzable using this suffix. Their existence justifies the fact that speakers may choose *-ano* as a variant of *-iano* to satisfy phonological requirements.

A study of similar suffixes in French and Italian allowed us to identify alternative strategies these languages adopt in order to satisfy the various constraints operating in each of them.

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Changes in inflectional class as a means to repair phonology

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One of the triggers of morphological variation is the emergence of phonological conflicts during the regular evolution of languages. This paper investigates changes in the inflectional class of the Catalan verbs as an alternative to phonological solutions to repair phonology.

In Catalan, loss of class II -Ē- thematic vowel, which derives from Latin third and second conjugations (PERDĒRE > *perdre*, SEDĒRE > *SEDĒRE > *seure*; cf. Davis&Napoli 1990), gave rise to consonant clusters that did not respect the phonotactics of the language (e.g., *sibilant+[□] contacts). In this case, different phonological repairs were activated resulting in dialectal variation, such as blocking of Ē-loss (general solution: CRESCERE > *créi.xer*), consonant insertion (southern Valencian: CONOSCERE > *coneix.tre*), or laxing of phonotactics (northeastern Catalan: CRESCERE > *crei.xre*). An alternative solution was the change in inflectional class (II > III in Alghero Catalan: CRESCERE > *crei.xir*).

An unnoticed case that further illustrates this situation comes from class I verbs of Balearic and Alghero Catalan. These varieties present the old ‘zero’ ending in 1st sing.-present-indicative forms, which derives from loss of Latin -Ō. Among these forms, those from verbs with roots ending in a vowel do not fit the regular prosodic profile of Catalan words, which do not end in vowel unless it is (part of) an inflectional or derivational morph. In this case, Balearic Catalan resorts to a phonological repair and adds a final glide (*evacuar*: *evacu[w]*, *variar*: *vari[j]*, *canviar*: *canvi[j]*). Alghero Catalan, instead, turned to a morphological solution and changed the conjugational class of these verbs: they migrated to subclasses of verbs with ‘extension’ (i.e., post-root sub-thematic affixes), which show this (consonant-ending) affix in 1st sing.-present-indicative forms, among others (*evacu-eig*, *vari-eig*, *canvi-c*, where *-eig* and *-c* are extensions of two subgroups that class I has in Alghero Catalan). Unless we notice the aforementioned prosodic problem, the fact that all verbs whose root ends in a vowel change sub-conjugation is seen as random analogy.

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Animacy in the Development of the Russian Predicate Adjective in the 19th and 20th Centuries

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The Russian language has undergone a number of morphological and morphosyntactic changes over the last two hundred years. Not least of these is the change in the predicate adjective, which can take three forms: short form, long form instrumental and long form nominative.

Previous corpus studies have examined the predicate adjective, with the copula verb *byt'* (to be) (Gustavsson 1977) and in specialist usage (Bazhenova 1992) and development over an extended period (Benson 1952). Our research concentrates on the changes over a 200 hundred year period, using a corpus of over 30 million words encompassing works of fiction, non-fiction and drama. The corpus was searched using a simple Concordance program, and the results analysed using a number of statistical methods in order to determine the relative factors and their significance to the predicate adjective choice.

In this poster presentation, we examine the use of these forms, the choices which determine them, and how these have changed. Copula verb, type of noun (concrete, abstract, animate, and so on) and temporality all play a role in determining the predicate adjective. We concentrate on the role of animacy of the noun in predicate adjective choice, and how this has changed in the period 1800-2000. In summary, statistically significant differences ($p < 0.05$) have been observed over this time period, particularly in terms of short form usage and variations in this usage across fiction, non-fiction and drama texts. This methodology may provide an opportunity to formulate a technique for predicting future changes.

This research forms part of the author's PhD research, which in turn is part of the Surrey Morphology Group's project *Short Term Morphosyntactic Change in Russian*, funded by the Arts and Humanities Research Board, UK.

Variation and change in morphology and syntax

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The paper argues for a polycentric view of language change: change can be motivated either externally (via contact) or internally (*pace* e.g. Milroy 1992:88, Thomason 2003:689, who equate change with contact), and, in the latter case, it can arise within any structural component (*pace* Longobardi 2001:278 and much current research in minimalism, claiming that syntax is never the primary locus of syntactic change). Within each component, as Ohala (1989) shows for phonology, ‘change is drawn from a pool of synchronic variation’.

To illustrate, I will draw examples from the empirical domain of Romance object-agreement phenomena. While past participle (= PtP) agreement with direct objects in perfective periphrastics was investigated extensively, the main focus of such investigation is on the syntax of (object-)agreement as well as – on the diachronic side – on its progressive reduction over time, eventually resulting in complete loss (as in Spanish or Romanian).

In this paper I will focus instead on (variation in) the morphology of object agreement, showing that changes affecting the inflectional *signantia* of agreement interact in some respects with changes in the syntactic rule but are in some other respects independent from it. In varieties such as French or Neapolitan, sound change eroded the inflections inherited from Latin, resulting in non-canonical (wrt. to Principle III of Corbett 2003, 2005) expression of agreement. While this fact was often held responsible for the retreat of (the syntactic rule of) agreement in French, it will be shown that this traditional view is unwarranted since the loss of agreement – as testified by cross-linguistic variation – proceeds independently on the syntactic and the morphological dimensions.

This is not to deny that there is an interplay between the two. In some dialects, in fact, this interplay even results in counterexamples to the principle of ‘morphology-free syntax’ (Zwicky 1996), as PtPs belonging to different morphological classes behave differently wrt. agreement. (Since this difference crucially depends on the PtP stem, such evidence militates in favour of a word-and-paradigm model of morphology.)

The search for data from Romance PtP agreement that may be of interest for a theory of morphology and morphological change will be concluded by an inventory of the options for the expression of non-agreement: this is most often syncretical with msg, more seldom with neutral agreement (in central Italian varieties which display neuter as a target gender, not just as a controller gender as opposed to Romanian, cf. Corbett 1991:152) and even more seldom relies on a dedicated exponent.

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Dialects in Contact and Inflectional Irregularity: The Case of Cassubian and Byelorussian

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A standard language may be based on a single regional dialect or it may be an interdialectal variety, thus levelling out structural features of different origin. Dialects as means of traditional oral communication are less exposed to standardization and literal culture and show a high degree of inflectional variation. On the other hand, morphological structures in dialects are more regular than the inflectional system of the standard language. In many cases, single instances of “unnatural” / irregular inflectional features in the standard language resist the impact of the more natural dialectal structures only by means of linguistic purification and imposing a low social prestige on the dialectal varieties, as has been attested not only once, e.g., in the case of Russian.

The concern of the present paper is a comparison of linguistic data from two regions, which have a “powerful” dialectal base. In the case of Cassubian and Byelorussian, a standard language has been established only in the course of the XXth century, and, by the way, standardization has been attempted at different times according to different local dialects. In the result inflectional variants never had been abolished completely from the standard languages – though the attested variation in the local varieties has been significantly reduced. What makes the situation even more complex is that most of the speakers in Cassubia as well as in Belarus are bilingual and have command of Polish or Russian respectively. The development of standard Cassubian and standard Byelorussian is thus influenced not only by regional dialects, but also by non-autochthonous contact languages.

In the present paper constellations of marker variation and morphonological alternations in Cassubian and Byelorussian will be examined with respect to their dialectal base and the foreign contact languages. The main question is if regular morphological structures from the dialectal base can enter the standard variety, even when the same structure is realized irregularly in the contact language Polish or Russian. There is incidence that they can, but the data allows for different interpretations. The thesis stands out that dialectal features tend to expand and standardize even against the influence of a prestigious contact language, if they contribute significantly to the regularity of an interdialectal inflectional system.

The form of mirror principle violations in Kĩtharaka

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Based on a detailed analysis of a previously unanalyzed Bantu language, Kĩtharaka, we aim to show that derivations strikingly similar to those needed to derive Greenberg's Universal 20 are also needed inside morphology. Traditionally, a curious asymmetry arose between morphology and syntax: morphology is governed by the mirror generalisation while (part of) syntax is governed by the less restrictive U20. Our results show that both morphology and syntax are uniformly governed by U20, and hence the kind of derivations needed for U20 (Cinque 2005, Abels & Neeleman 2007) are also needed word-internally. Such derivations are needed to account for the simplest scope facts between the various morphemes of the Kĩtharaka verb, realized at PF as in (1). (<= precedes, Rec = reciprocal, apl = applicative, stat = stative, R = verb root).

- (1) R < Rec1 outer cause < apl1 < Rec2 < able < erratic < hab < apl2 < stat
I II III IV V VI VII VIII IX X

Although *apl* 1 & 2 follow outer cause (III), a causative morpheme attaching to transitives and robustly triggering coercive force, the applicative can introduce a DP for whose benefit the caused event is done, or on whose behalf causation is done. By the MP, the former reading is expected to have one of the applicatives precede outer cause, an impossible ordering. We have an MP violation here.

Rec1 allows for coindexation between the embedded external argument and the direct object, as expected by the MP. However this same coindexation is allowed when there is *outer cause* and *Rec2*, contra the MP expectation. To account for this violation, we assume that *Rec 2* can sometimes be a copy of *Rec1*.

Able (VI) scopes over *outer cause* (III), and so does *erratic* (VII), *habitual* (VII), and the *stative* (X), in conformity with the MP. Between *outer cause* and the *stative* however, there exist MP violations. Although *able* precedes *erratic*, it scopes above *erratic*. *able* however scopes below *hab*, as expected under MP.

Consider lastly *hab* (VIII) and *stat* (X). These two cannot co-occur on the same verb. However in a biverbal construction with an expletive BE-verb, a kind of DO-support *hab* must occur on the higher verb, and *stat* on the lexical verb. In a bottom-up model of phrase structure, *hab* must be higher than *stat* in the base, another MP violation. Combined, these factors give us the base order in (2).

- (2) hab > apl2 > stat > able > erratic > Rec2 > outer cause > apl1 > Rec1 > R
VIII IX X VI VII V III IV II I

Deriving the surface order: (i) The root rolls past *Rec1* (ii) root, *Rec1* cycles past *apl1* and *outer cause* (iii) root, *Rec1*, *outer cause*, *apl1* rolls past *Rec2* (iv) root, *Rec1*, *outer cause*, *apl1*, *Rec2* cyclically moves past *erratic* and *able*, finally (v) root, *Rec1*, *outer cause*, *apl1*, *Rec2*, *able*, *erratic* cycles past *stat*, *apl 2* and *hab* giving the surface order root, *Rec1*, *outer cause*, *apl1*, *Rec2*, *able*, *erratic*, *hab*, *apl2*, *stat*.

Semantic variations of suffixed nouns in dictionaries and Web corpora.

The case of French deverbal *-oir* nouns.

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It is usually assumed that constructed neologisms semantically differ strongly from those nouns that have been stored in dictionary for a long time. We will show that despite superficial differences in meaning, French *-oir* suffixed nouns (*N-oir*) share fundamental semantic properties, be these nouns stored in dictionaries or collected from the Internet.

Our study is based upon an *N-oir* list attested in a multi-volume French dictionary: TLF (*N-oir*_{TLF} corpus), as well as upon a close analysis of *N-oir* neologisms collected on the Web (*N-oir*_{W3} corpus). Both corpora constitute an original set, since they reveal the great productivity of the V to N *-oir* Formation Rule: 400 nouns in *N-oir*_{TLF}, and 500 different nouns in *N-oir*_{W3}.

On one hand, as expected, *N-oir*_{W3} show peculiar properties: they are often motivated by discourse grounds (whereas *N-oir*_{TLF} fulfill denotation function); they may be “re-creations” or belong to sublanguages or non standard speech registers.

However, on the other hand, *N-oir*_{W3} and *N-oir*_{TLF} present important semantic similarities. As traditional analyses account for, both dictionary and Web corpora *N-oir* nouns are mainly interpreted as instruments (GRATTOIR - *scrape-oir*=‘*scraper*’) or locations (FUMOIR - *smoke-oir*=‘*smoking room*’). Moreover, as predicted for French deverbal nouns in general (Plénat (2005), Hathout & al. (2003), Roché (2003)), both *N-oir*_{TLF} and *N-oir*_{W3} may be interpreted in relation to any participant of the base verb, or to the base verb process itself, as soon as the morphological rule at play is not explicitly constrained: location¹ (FUMOIR; GLANDOIR - *bum-around-er*= ‘*bum around place*’), instrument (GRATTOIR; RENIFLOIR - *sniff-oir* = ‘*nose*’), agent (BASSINOIRE - *bore-oir*=‘*boring person*’); CONSOLOIR - *console-oir*=‘*consoling person*’), patient (TIROIR - *draw-oir*=‘*drawer*’; ASTICOTOIR - *needle-oir*=‘*needled person*’), process (INTERROGATOIRE - *question-oir*=‘*interrogation*’; DÉBALLOIR - *pour-out-oir*=‘*pouring-out*’).

Furthermore, we will show that the base V semantic properties seem to be the same for *N-oir*_{TLF} and *N-oir*_{W3} neologisms: these V instantiate predicates of all possible aspectual types (Vendler (1967), Smith (1990), Foley & Van Valin (2002)), mainly dynamic ones, as only a few stative predicates have been met (AIMOIR - *love-oir*, SOUFFROIR *suffer-oir*, VIVOTOIR – *struggle-along-oir*): activities (PARLOIR - *speak-oir*, TRAVAILLOIR, - *work-oir*), accomplishments (DOROIR - *gild-oir*, ASSASSINOIR - *murder-oir*), semelfactive (PÊTOIRE - *burstf-oir*, SURSAUTOIR - *jump-oir*), achievements (ALLUMOIR – *put-on-oir*, ARRIVOIR - *arrive-oir*).

All in all, we will see that the same LFR can account for both (long time) attested *N-oir*_{TLF} and just coined *N-oir*_{W3}

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¹ Underlined nouns come from *Noir*_{W3} corpus; the other ones, from *Noir*_{TLFi} corpus.

The Classification of Nouns in German: Four Genders and Lexical Proper Names

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Linguists classify nouns in order to explain their different linguistic properties. The concept of inflectional classes is used for the explanation of the inflectional behavior of nouns while gender is used for the explanation of their syntactic behavior. The two concepts are related to each other in that inflectional classes are in part determined by gender. Interestingly, Eisenberg's (2000) suggestion to regard the weak masculines as a distinct gender in German allows doing away with the concept of inflectional classes. Under this perspective, normal masculines and neuters behave alike in singular inflection while feminines on the one hand and weak masculines on the other constitute two other relevant groups. With respect to their external syntax, weak masculines and normal masculines behave alike.

However, a subgroup of proper names has properties that do not fit in this pattern. I will call these proper names having the feature of being allowed to appear in syntax without a definite article 'lexical proper name'. While the genitive of appellative feminines is identical to the nominative singular, feminine lexical proper names in prenominal position have the suffix *-s* (e.g. the female Christian name *Karin* with the prenominal genitive *Karins*). Moreover, while the genitive of many appellative masculines has both a short and a long form in stylistic variation (e.g. *Horsts* and *Horstes* 'nest (gen.)'), masculine lexical proper names have only a short form (e.g. *Horsts* but not **Horstes*). In other syntactic positions, the genitive of lexical proper names has a zero suffix. Thus, the genitive of these nouns shows syntactically conditioned variation.

The inflection of lexical proper names obviously does not depend on gender information. This is noteworthy given that there are some proper names that seem to have no gender at all, namely names of companies like *Opel* or *Siemens* that may never be used with an article.

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Why are Object-Verb-*er* compounds so difficult for English-speaking children?

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English-speaking children do not produce novel Object-Verb-*er* (OV-*er*) compounds (such as *floor sweeper*) until around five years (e.g., Clark, Hecht, & Mulford, 1986). However, children can use the morpheme *-er* correctly from three years (Clark & Hecht, 1982) and can reliably differentiate head and modifier within other compounds by two years (Clark, Gelman, & Lane, 1985).

Clark (e.g., 2003), noticing children's errors were sometimes object-verb reversals (e.g., *sweep-floor* or *sweeping-floor*), attributed children's difficulty to the high frequency of phrasal verb-object order.

To test that explanation, we asked English-speaking children to produce two kinds of novel lexical constructions, both of which reverse typical phrasal ordering and are equally frequent in the input: Verb-*ing* Noun (like *shooting star*) and OV-*er*. We found that children produced far more target Verb-*ing* Noun forms than OV-*er*. When the children did not produce the target form, they often included the target morpheme in their construction. For OV-*er* targets, when they used *-ing*, it was most often to create a Verb-*ing* Noun (such as, *sweeping man*) rather than a Verb-*ing* Object (such as *sweeping-floor*) construction.

These results suggest that it is not reversal of phrasal ordering that make OV-*er* constructions difficult for children. We argue that the difficulty may be the semantic relation between the two words within the compound. Preschool children have been found to have difficulty in interpreting noun-noun compounds with semantic relations that refer to function (e.g., *bottle brush*). Similarly, children must understand OV-*er* compounds in terms of their function (e.g., a *dishwasher* is FOR washing dishes). Because the function cannot necessarily be observed quickly, children may have a harder time learning the meaning of OV-*er* compounds than constructions like Verb-*ing* Noun. Verb-*ing* Noun can have a variety of semantic relations between the two words, some of which may be simpler for children to learn over a short period of observation. We discuss some crucial tests of our interpretation.

Variation in Tilapa Otomi Verb Conjugations

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Tilapa Otomi is a Mesoamerican (Otopamean, Otomanguan) language on the verge of extinction spoken by a dozen of old speakers in the village of Santiago Tilapa, Mexico State, Mexico. This language is a linguistic island with respect to other Otomi languages as it shows idiosyncratic phonological developments and preserves much of the complex morphological paradigms from Old Otomi (Cárceres 1580/1907), which have been largely reanalysed in other languages of the family.

In this paper, I show an interesting case of morpho-lexical variation in the selection of TAM inflection by verbs. Tilapa Otomi verbs inflect for more than 17 TAM categories by means of complex series of proclitics, at times combined with prefixes. Such markers also cross-reference subject. Example (1) shows Series A and D of Centripetal Present:

- (1) a. **bá=**juradi
3.CENTRIPETAL.PRESENT(A-Series)=run
'He's running hither'.
b. **bá=ti-**nömi
3.CENTRIPETAL.PRESENT(D-Series)=PREFIX-water
'He's watering (the field) hither'.

Verbs are arrayed in three conjugations attending to the TAM series they select in eight critical paradigms (Centripetal Present; Activity Imperfect; Past; Perfect; Plu-perfect; Present Irrealis; Past Irrealis and Future). Whereas the conjugations may be discerned from a number of verbs that behave as a unified class for each case, the variation within each class is outstanding. A common verb –save the exemplar ones– may take six of the paradigms in a regular way, and behave like a verb from a different conjugation with respect to the other two.

Taken as a whole, the TAM inflectional system in this language is organized in lexical trends rather than absolutes, implying that the speaker must employ great lexical memory in this respect. The phenomenon may be explained as motivated because of the social pressure of a language close to extinction, as shown in Dorian (1994), but it may just reflect an idiosyncratic characteristic of Otomanguan languages, as extensive irregularity has also been reported in other strongly spoken languages, such as Chinantec (Foris 2000) and Amuzgo (Smith-Stark & Tapia García 2002), among others.

In the paper, I describe the conjugations in detail starting from typical cases and then I show the different types of variation patterns I have found in a sample of more than 150 verbs of which I have a complete paradigm.

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Optional Multiple Plural Marking in Maay

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This paper shows how a previously undocumented phenomenon in Lower Jubba Maay (LJM; Cushitic; Somalia) supports a claim of Optimal Construction Morphology (OCM; Inkelas et al 2006).

C(onsonant)- and V(owel)-final nouns take different plural marking in LJM. V-final roots take *-yal* (1a); C-final roots have 3 plurals: *-o*, *-yal*, and *-o-yal* (1b).

- (1) a. buundo-**yal** ‘bridges’ bakaile-**yal** ‘rabbits’ mindi-**yal** ‘knives’ liwa-**yal** ‘lions’
b. basal-**o** ~ basal-**yal** ~ basal-**o-yal** ‘onions’ eey-**o** ~ eey-**yal** ~ eey-**o-yal** ‘dogs’
yahas-**o** ~ yahas-**yal** ~ yahas-**o-yal** ‘crocodiles’ af-**o** ~ af-**yal** ~ af-**o-yal** ‘mouths’

Unlike Breton double plurals (Anderson 1986) the LJM plurals are semantically identical. This is a problem because M(ultiple) E(xponence) is blocked in most models by rule blocks, UNIQUENESS, or ECONOMY (Anderson 1986, Noyer 1993, Kiparsky 2005) and is even claimed not to exist (Anderson 1992, Noyer 1993).

I analyze LJM in OCM, a subcategorization- and constraint-based model. Below are subcat frames for the plural markers:

- (2) *-o* suffix: [[C#]_{Noun Root} *-o*]_{Stem}
 -yal suffix: [[]_{Noun} *-yal*]_{Word}

-o attaches only to C-final Roots, so V-final nouns take only *-yal* and **-yal-o* is ruled out. We now must explain optional ME on C-final roots. ME is driven by BE-WORD, a constraint violated twice by Roots and once by Stems, giving preference to affixation that advances a form towards wordhood. ME occurs when BE-WD outranks ECONOMY (‘The fewer affixes the better’; Noyer 1993:17). Due to free constraint ranking (Kager 1999:406) OCM also predicts *optional* ME, attested in LJM.

Permuted rankings of the constraints yield all and only the attested plurals: Stem-level rankings BE-WD >> ECON and ECON >> BE-WD select *-o* and *-Ø* respectively; these are inputs to Word level. With input *-Ø* both Word-level rankings select *-yal*; with input *-o* the rankings select *-o-yal* and *-o* respectively. Thus I account for optional ME morphologically with no need for phonological, syntactic, or semantic explanations.

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Case Study on the Acquisition of Slovenian Noun Inflection

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This paper in the framework of Natural Linguistics (cf. Dressler 1997) presents results of a longitudinal investigation of the language acquisition process in a Slovenian child (*Petra* – from 1;0 to 4;0) and aims to contribute to the general question of how the morphology typology of the target language influences the acquisition process. The language materials used as evidence consist of language data from studies of other linguists and of language data gathered in own empirical investigations of first language acquisition (ie. transcriptions of video recordings for statistical comparison and a developmental diary for the detection of the first uses of new categories). – Two of the four main Slovenian noun declension classes are acquired earlier. The first feminine nouns of the first declension class (ending in *-a*) are among the first words (beginning with 1;0 – *mama* “mammy”, *gaga* “duck”, *voda* “water”) due to the productivity of the paradigm and phonological reasons (most natural vowel). The first masculine nouns typically ending in a consonant (consisting of only one syllable and therefore deviating from the preferred trochaic syllable pattern) arise much later (beginning with 1;11 – *noj* “ostrich”, *zmaj* “dragon”, *gol* “goal”) and coincides with the first uses of numerals, plural noun forms, verb imperatives and gender distinction. Before the age of 1;11 the Slovenian child uses the hypocoristic noun ending in *-i* to refer to feminine or masculine persons (*mami*, *ati*) and domestic animals (*dodi* dog “Lori”). Feminine nouns ending in a consonant (*miš* “mouse”, *luč* “light, lamp”, *rit* “bottom”) and being part of the second main declension class of Slovenian nouns and neuter nouns forming the fourth main declension class of Slovenian nouns (*milo* “soap”, Pl. *vrata* “door”, *vino* “wine”) are not preferred by the child (with 2;4 neuter *vedro* → *vedo* “bucket”). Therefore in the age of 2;0 and afterwards, she uses diminutive noun forms instead (*miš-ka*, *luč-ka*) or uses the typical endings of the first (feminine) or third (masculine) declension class (fem. *rit* → *rit-a*; neuter Sg. *milo* → *mil*, neuter Pl. *vrata* → *vot-e*, *vino* → *vin*). This shows the preference of the child to group all nouns into two distinct and productive paradigms (feminine nouns ending in *-a* vs. masculine nouns ending in a consonant). – In Slovenian it is common to use diminutive nouns to denote pragmatic meanings (ie. familiarity, closeness, affinity) and/or smallness of the referent (ie. *fant* → *fant-ek* “”, *fant-ič-ek* “(little) boy”, *sestra* → *sestr-ica*, *sestr-ič-ica* “(little) sister”). The Slovenian child under investigation develops diminutive nouns mainly after the acquisition of fundamental noun and verb paradigms in the protomorphological period (especially nominative and accusative singular and plural of feminine and masculine nouns of the first and third declension class; present, past and future tense forms of verbs of the most productive 1st macroclass of Slovenian verbs in the singular, plural and dual), but much earlier than other derivative nouns in Slovenian (diminutives in the age of 2;6: *zmaj* → *zmajček* “toy dragon”; *piščanec* → *piščanček* “chicken”; *voda* → *vodica* “water”; *mama* → *mamica* “mother, mommy”; *medved* → *med(ved)ek* “bear”; *duh* → *duhec* “ghost”; in the age of 3;2 *punca* “girl” → *punč-ka* “girl, doll”; *rit* “bottom” → *rit-ka* “bottom (of girls)”; *pljunek* → *pljunk-ec* “spit”; 3;3 *korenje* → *korenj-ček* “carrot”). Only hypocoristics ending in *-i* appear in the premorphological period. In the first stage of acquisition diminutive nouns and their corresponding basic nouns often exist side by side without a clear-cut difference in meaning. Diminutives are associated with pragmatic meanings since children acquire meanings contextually. The concept of smallness is not necessarily part of the noun meaning, but later on the pragmatic meanings and the semantic meaning of smallness are often combined.

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The Variability of Compound Stress in English: Rules or Exemplars?

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In both psycholinguistic and theoretical-linguistic circles there is a debate about the nature and role of symbolic rules, associative networks and analogical or exemplar-based models in the organization of language in general, and of morphology in particular (see, for example, Skousen et al. 2002). Compounding has featured prominently in this debate (cf. e.g. the papers by Krott and colleagues), and the present paper complements this line research by focussing on an aspect of semi-regularity in compounds that has received little attention so far: stress assignment in English noun-noun combinations.

English noun-noun compounds are traditionally assumed to be subject to the Compound Stress Rule, which states that the left-hand constituent is more prominent than the right-hand constituent. However, not all English noun-noun compounds abide to the Compound Stress Rule (cf. for example, *ópera glasses*, *wáitch-maker* and *Óxford Street* versus *steel brídge*, *morning páper*, and *Madison Ávenue*). Rightward stress is far from exceptional, and the nature of the observable variability is still rather unclear (but see Plag et al. 2007).

In this paper we will use two large corpora of noun-noun compounds (based on CELEX and the Boston University Radio Speech Corpus, more than 9,000 items overall) to test categorical, rule-based approaches and exemplar-based approaches with regard to the problem of variable compound stress. Exemplars are stored experiences of previously encountered linguistic items. Exemplar-based models now assume that grammar emerges from individual exemplars, rather than being the effect of an independent system of rules or constraints that operates on lexical items. Specifically then, for compound stress the basic hypothesis is that each new compound is assigned the stress pattern of the majority of those exemplars that are most similar to the new compound.

In this paper we will use two current implementations of exemplar-based algorithms, TiMBL and AM::Parallel. All compounds in our corpora were coded in terms of the semantic and structural features that are traditionally held to influence stress assignment, and both kinds of model were given these features as information source. Additionally, the exemplar-based models were able to use the constituent family of each compound, a factor that has been shown to be influential with variable semantic and morphological properties of compounds (e.g. Gagné 2001, Krott et al. 2002). A ‘constituent family’ is the set of compounds that share the first (or the second) constituent with a given compound.

We will show that the exemplar-based models grossly outperform the rule-based implementations. Furthermore, both TiMBL and AM reach best classification results if the model is fed only with constituent family as an information source. These results challenge rule-based approaches to compound stress, but are in line with recent findings concerning the role of constituent families in related studies of compound semi-regularity (see the references above). Furthermore, our implementations demonstrate that the computation of stress in compounds does not necessarily require reference to abstract structural and semantic features.

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Order in Sanskrit Preverbs

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In Classical Sanskrit, numerous verbal prefixes ('preverbs') combine with verbs to give new meanings (gam 'go', versus sam+gam 'accompany'). The phenomenon is not unique to Sanskrit; many other Indo-European languages have comparable structures, with Celtic and Slavic being notorious for multiple instances of these elements.

Classical Sanskrit also shows multiple preverb stacking. While in theory there are hundreds of possible combinations of these stacked preverbs, only some of these combinations actually appear. In fact, some preverbs show consistent trends in positioning relative to the verb.

I investigate possible reasons for these positioning trends, including relevant phonetic and semantic factors. While phonetic environment could potentially play a role in preverb ordering, it does not appear to play a significant role.

Semantics has long been considered a primary motivation for preverb ordering in Sanskrit. In discussing preverbs in his Sanskrit Grammar, Whitney makes the claim that the order of preverbs is determined by the meaning, which is compositional (such that the meaning of X+Y+Z+VERB, is analyzed as VERB, then Z+VERB, then Y+[Z+VERB], etc.). Contrary to this, I show that while semantics may play some part in ordering, it is not the only motivating factor.

What emerges from this investigation, therefore, is that there are lexically particular ordering specifications associated with Sanskrit preverbs both in the Classical language and also in Vedic Sanskrit. I show moreover that these lexical particularities are likely to be inheritances from Proto-Indo-European, since a comparison of the Sanskrit situation with what is known about preverb ordering in other Indo-European languages, particularly Celtic and Slavic, shows cognate forms with similar lexical specifications.

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**Parsimony and efficacy in language:
A psycholinguistic analysis of the dual *binyan* system of Hebrew**

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This talk analyzes the *binyan* verb-pattern system of Hebrew as a window on questions of parsimony and efficacy in linguistic processing and acquisition. In structural linguistics, efficiency in language means parsimony of theory (*Occam's Razor*), in the sense that "what can be explained by the assumption of fewer things is vainly explained by the assumption of more things" (Boehner, 1957). *Economy of derivation and representation* – that grammatical structures must exist for a purpose – assumes that principles of economy determine the shortest linear road from logical form to syntax (Chomsky, 1995; Fox, 1999). When debating between accounts of linguistic phenomena, phonologists (for example) evoke parsimony as the determining principle.

However this reductionist, digital account of how language operates is not easily reconciled with the analogue nature of human experience and language. New, neural-network stochastic accounts of language acquisition and processing based on pattern detection and generalization claim that efficient processes in brain and cognition do not necessarily operate parsimoniously (Bybee, 2006; Elman, 2005) and show how complex linguistic architectural organization emerges from simple beginnings (McClelland & Patterson, 2002; Tomasello, 2003). Under this view, linguistic integration is a more parsimonious process in the human application of easiest path adaptability through interconnected systems.

The Hebrew *binyan* system expresses transitivity values in seven non-concatenative patterns which organize the Hebrew verb lexicon and determine the verb-argument structure of the Hebrew sentence. Thus the application of *binyan* patterns to root *s-g-r* yields basic *sagar* 'shut', passive / inchoative *nisgar* 'close / be closed, Int.', causative *hisgir* 'extradite', passive *husgar* 'be extradited', and reflexive *histager* 'close oneself'.

This talk will employ a psycholinguistic perspective to show that the Hebrew *binyan* system actually consists of two parallel, redundant subsystems, each expressing the full set of valency functions – as shown for low-transitivity *Nif'al nixnas* 'enter' and *Hitpa'el hitkanes* 'gather', high-transitivity *Pi'el kines* 'gather' and *Hif'il hixnis* 'insert'. The redundancy of the system provides the platform for creating the richly complex verb architecture of Hebrew. The two subsystems are integrated via a network of weak links and partial regularities (Landauer & Dumais, 1997), which enable easy and early acquisition (Berman, 1993). In search of "how perfect language can be" (Chomsky, 2002), efficacy in learning and processing an extremely complex system does not necessarily arise through classical parsimony.

Athapaskan Middles

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Middle voice has received extensive discussion in the Athapaskan literature (e.g., Arce-Arenales, Axelrod, and Fox 1994, Kibrik 1996, Rice 2000a, b, Thompson 1989, 1996), as well as in the literature more generally (e.g., Kemmer 1993, Kaufman 2007). Middles are overtly marked in Athapaskan languages, with middle marking occurring in a wide range of constructions, including reflexive, reciprocal, incorporated body parts, self-benefactives, intransitive iteratives (go and return). These represent prototypical middles, as discussed in the literature.

In addition to these uses of middle voice, there are other constructions in which it regularly occurs in some Athapaskan languages. In particular, it is found in the so-called errative construction (doing an activity referred to by the verb excessively or incorrectly and being unable to stop or escape from the consequences), in the perambulative construction (Verb around), and in the activity incorporate construction (do X while Y-ing).

What do these various constructions have in common that leads to middle marking? I propose that middle marking is used when two elements are present that refer to one. These two elements may be entities, as in the cases of reflexive, reciprocal, self-benefactive, and body part incorporate constructions. The two elements may be locations, as in the intransitive iteratives, where the source and goal are identical. They elements may also be identical in terms of time, as in the activity incorporate construction, and may involve identical events as well, as in the errative and perambulative constructions, where the event specified by the stem is repeated.

The use of middle voice marking in these languages has several implications. First, it suggests that middle voice in these languages is not best identified as argument decreasing (e.g., Kibrik 1996), given that it is used in constructions involving events as well as those involving entities. Second, it suggests that middles truly involve voice and not syntactic valence. Third, more generally, it suggests that middle voice in its most general form involves two distinct elements, be they entities, events, spaces, or times, and what is core to middles in these languages is that these two elements map onto one. An adequate theory of middles must allow for these unusual cases of middles involving times.

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The morphology and semantics of roman letter words in Chinese

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Chinese is showing the effects of internationalization through contact with English. One new trend is the use of abbreviations (*WTO*) and *hybrid words* (*X 光* ‘X-ray’). This study surveys hybrid words in the LDC’s *Chinese Gigaword 3rd Edn.* corpus. We explore relationships between the morphological and semantic structure of hybrid words and words written entirely with Chinese characters. The orthography of hybrid words belies their Chinese structure: “foreign” letters are used as a new type of Chinese *zì* (syllable-morpheme-character) that we call *roman letter zì*. These new *zì* combine with native *zì* to create hybrid compounds, which are formed with the same morphological processes as native compounds.

Most hybrid words are endocentric nominal compounds of the form [N1 N2]_N. The head (N2) is a Chinese *zì* constituent, and the modifier (N1) is a letter *zì* constituent (e.g. *X 光*). The roman letter modifier expresses a new concept, while the Chinese head expresses the word’s core lexical and semantic category. This pattern is preferred because roman letter constituents are unfamiliar and opaque and therefore function more appropriately as modifiers. This structure also positions hybrid words in Chinese type hierarchies, facilitating their interpretation in terms of familiar semantic categories. The head ‘sinicizes’ the hybrid word by serving as its hypernym.

We conclude that hybrid words create a paradox in Chinese. They contradict the longstanding tradition that “(foreign concepts) must be made Chinese in form” (Pasierbsky 1989:102). At the same time, hybrid words *are* sinicized since their structure is the same as that of native compounds. Hybrid words challenge notions of what is native in Chinese and reveal the flexibility of compounding in incorporating new elements into the language.

Word creation: definition - function – typology

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In this paper, the term "word creation" refers to all techniques for the production of new lexemes, which are not covered by the productive rules (or models) of regular word formation. It includes not only the coining of entirely new words not based on any previously existing linguistic elements (German *Urschöpfung*), but also extragrammatical operations such as shortening or blending lexemes or phrases. The creative character of these operations is justified by the fact that their output cannot be predicted by the input and a given rule or model, but, in turn, the operation is devised according to a desired output and an available input. As a consequence, the output of an extragrammatical operation is less transparent than its source form (e.g., of a shortening) or than the output of a corresponding grammatical operation (e.g., of a blend as compared to a regular compound). This reduction of transparency is one of the primary functions of word creation, the other one being the production of sound shapes with special characteristics such as, e.g., a specific rhythmical contour. Typical fields of application are different kinds of (mostly humorous) literary texts, brand names and others.

The paper will contain a complete typology of creative techniques, which integrates several existing partial typologies such as Algeo 1978, Cannon 1986 and 1989, Grésillon 1984, Kobler-Trill 1994, Soudek 1978, Steinhauer 2000 and others, trying to overcome certain shortcomings and inconsistencies. It has been developed and thoroughly tested as part of a long-term project on the history of German brand names (Ronneberger-Sibold 2004). However, it can be and has been applied to word creations in other fields as well.

A promising subject for further research is the application of the typology to other languages in order to test in what ways it needs to be modified under different inner- and extralinguistic conditions, such as different stress patterns, morphological structures, writing systems, language policies, etc. First results obtained for French suggest an interesting interplay of universal and language specific features in word creation.

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Russian animacy: variation and its trends

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Animacy in Russian is thought to be a category with dominant semantic kernel and narrow semantically opaque periphery (Corbett, Timberlake). This conception is however wrong. One of the numerous phenomena within the peripheral zone of this category are those units which emerge in metaphoric usages if an originally animate noun (or longer nominal expression) develops such a meaning in which it denotes an inanimate object, or the other way round. A study based on the National Corpus of the Russian Language revealed that these units demonstrate significant variation in morphological setting. It was also found that such units drift similarly over time: they start with marking animacy according to their original meaning but gradually prototypic relations between form and meaning are being established. The distribution of animate and inanimate models at a particular moment of time is chiefly determined by the distance covered by a particular unit in such a development. Thus, the ratio of the animate model is higher with *belyj vorotnichok* (lit. ‘white collar’, that is ‘white collar worker’), than with *sinij vorotnichok* ‘blue collar’, since the former idiom is much older. Besides, the speed of developments just described depends on the position of the object denoted in the metaphoric use on the animacy hierarchy.

An experiment was held in order to observe speakers’ behavior in a picture-story task: animate objects were designated by inanimate nouns and vice versa. The subjects passed through several stages: they started with the lexically determined morphological setting, then followed a stage of hesitations, finally the speakers stuck to one of the two models.

Results. 1. The genuine nature of the category at issue reveals itself through peculiarities of variation and shifts in particular nominal units. 2. Animacy in Russian demonstrates properties of a classificatory category for nouns, but at the same time it has latent inflectional potential, which is awoken under certain circumstances.

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Is any change in word-formation word-formation change?

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This paper explores diachronic change in word-formation. The phenomena of change usually attributed to the domain of word-formation range from phonological alterations of affixes, grammaticalization and the extraction of word-formation elements out of loans to the extinction of word-formation patterns due to the loss of morphological productivity. But actually, these phenomena affect different levels of word-formation, i.e. word-formation products, word-formation patterns and the word formation system as a whole. Some of them are not even restricted to word-formation but do apply to other components of grammar as well. In my talk, I will show that most phenomena of change attributed to word-formation may be explained in terms of other types of change, e.g. phonological or semantical change.

I will argue that the only kind of change that is specific to the domain of word-formation is the change of word-formation constraints. This change of constraints is reflected in changes of morphological productivity. Therefore, word-formation change can be evaluated empirically.

So far, only little reference has been made to productivity in historical word-formation. Word-formation change has mostly been perceived as change in the word-formation system resulting from some extramorphological, e.g. phonological or syntactic process (Munske 2002). In his theory of word-formation change, Munske emphasizes changes affecting the word-formation inventory, namely the emergence and loss of word-formation means as well as formal and semantic changes in existing patterns.

As I will argue, studying the change of productivity for individual word-formation patterns can lead to a better understanding of changes in the word-formation system as a whole. Productivity may not only be helpful to understand why constraints on individual patterns vary diachronically. It may also help to explain changes in the word-formation system as changes in the word-formation system mainly result from the emergence and disappearance of word-formation patterns. The very first stage of analogical coining can be explained as a new pattern becoming productive, whereas for vanishing patterns, productivity converges to zero. My conclusion, therefore, will be that word-formation change can be defined in terms of change of productivity with the change of productivity dependening on constraints on the input or output of the word-formation pattern.

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Variation of Naming Procedures in German and Dutch

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In morphological theory, it is widely assumed that compounds have a classificatory meaning and therefore a naming function (e.g. Jespersen 1942, Zimmer 1971, Bauer 1988). However, despite the fact that compounding is a general device for creating “names” and that (nominal) compounding is highly productive in Germanic languages like German and Dutch, it is by no means the case that all nameworthy referents are denoted by a compound. Looking at A+N-compounds in German and Dutch shows that they are sometimes blocked by lexicalized A+N-phrases with a specialized meaning. For example, due to the existence of German *grüne Welle* (‘progressive signal system’) and Dutch *donkere kamer* (‘dark room’) we cannot form *Grünwelle* and *donkerkamer*. Compounds and lexicalized phrases do coexist both in German and in Dutch, although lexicalized phrases are more frequent in Dutch than in German.

Two questions arise from this observation. First, why is there variation in the realization of the naming function? Phrases are generally thought to be part of syntax, describing rather than naming entities. In contrast, compounds, as naming entities, are part of wordformation. Now in German and Dutch, contrary to English, phrases can be clearly distinguished from compounds by means of word stress, inflection of the adjective, and spelling. Lexicalized phrases that name entities thus break through this dichotomy of form and function. So why are phrases used as naming entities at all?

The second question is: What determines the choice between the two alternatives? We argue that there are several factors which have an effect on the choice between compound and phrase. First of all, we will show that analogy plays an important role when new names are coined. Also, it will be claimed that the loss of inflection in Dutch can be held responsible for the fact that lexicalized A+N-phrases are even more frequent in Dutch than in German. As to semantics, we will argue that the degree of metaphoric meaning plays an important role, too, for the question whether a nameworthy referent is designated by a compound or a phrase, as more metaphoric meaning tends to be realized as a phrase, at least in German. It will be shown that these three factors can be held responsible for the divergent development of the realization of the naming function in German and Dutch.

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The Stress of Words Suffixed with +i in Modern Hebrew

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Modern Hebrew includes some morphemes which although follow the same phonetic structure, the addition of which has caused variations in stem stress. This paper will examine one of these suffixes, the suffix +i, and demonstrate the factors that determine stress variability. The morphemes represented by the suffix +i are: 1. the second person feminine singular in the future tense and in the imperative, e.g. *titen-titni* 'you will give/2sg.m-f'; 2. first person noun inflections (possessive pronouns), prepositions and some adverbs, as well as verbs (accusative pronouns), e.g. *dod-dodi* 'uncle-my uncle,' *mul-muli* 'against-against me'; 3. gentilic substantives (indicating place of origin or religion), e.g. *yisra'el-yisre'eli* 'Israel-Israeli'; 4. adjectives, e.g. *yesod-yesodi* 'basis-basic'; 5. affectionate and diminutive nouns, e.g. *xamud-xamúdi* 'cute-cutie.'

Morphemes of type 1 and 2 are inflectional while those of type 3, 4 and 5 are derivational. The unmarked stress in groups 1-4 is ultimate in words with suffixed +i, whereas affectionate and diminutive nouns from group 5 are always stressed penultimately. Various factors determine the penultimate stress in words with suffixed +i morphemes of type 1-4. They can be morphological and phonological in inflection, or lexically marked as foreign in derivation.

The morphophonemic role of words with the suffix +i protrudes in inflection. High vowels in the last syllable of the stem and phonetic two-consonantal roots determine the penultimate stress of inflected verbs for the second person feminine form in the future tense and the imperative. Verbs are always penultimate stressed with the first person accusative suffix +i which is realized as [+ni]. This unstressed [+ni] allomorph also occurs in a few lexicalized adverbs and prepositions.

Penultimate stress in derivational adjectives with the suffix +i is dependent on loan word origin. Gentilic substantives with the +i suffix are also penultimately stressed when their origin is derived from loan names. Ultimately stressed gentilic substantives are not very common, most of these originating in the language prior to the development of Modern Hebrew or at the early stages of its current formation. All these relatively few derived nouns and adjectives need to be lexically marked for penultimate stress assignment. It seems that the penultimate stress of the affectionate and diminutive nouns is also a foreign influence, as this word formation device is new in Modern Hebrew.

How to do things with moras: variation and change of quantity alternations across Upper German dialects

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1. The problem: In East Franconian (Heilig 1898) long stem vowels of singular nouns correspond to short vowels in the plural: *ti:/* ‘table’, *ti/* ‘tables’. Whereas at first sight it looks as if the vowel were shortened in the plural, a systematic inspection of other contexts reveals that it is rather the singular form where the vowel is lengthened (which is, of course, highly countericonic).

2. Historical and comparative background: A comparison with Bavarian and Alemannic shows that the East Franconian pattern results from a series of prosodic changes – monosyllabic lengthening, apocope, degemination – whereby the latter two obscured the original environments for lengthening. Also, the lack of lengthening in the plural turns out to be exceptional.

3. Analysis: Auer (1989) proposes for Bavarian that the plural morpheme of the relevant inflectional class is an unpronounced mora whose addition makes the plural behave like a disyllabic form. After a discussion of this approach and its applicability to East Franconian we will conclude that the proposal is technically feasible, must indeed follow directly from the assumptions made by moraic theory, and, as a further advantage, lets appear plural formation much more iconic – however, at the expense of introducing *ad hoc* the category of silent moras. How can the proposal be further supported? Kiparsky (1991) introduces the idea of catalexis, i.e. prosodic structure without association to segmental content. As an instance of catalexis, East Franconian plural formation is embedded in a much broader crosslinguistic and theoretical context. Furthermore, if there are catalectic moras as plural morphemes in some dialects, then we might expect ordinary moras to occur as plural morphemes in others since catalexis is the typologically more marked option. This is exactly what we find in the Alemannic dialect of Obwalden.

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Typological Effects on the Mental Lexicon: The Case of Hebrew derived nouns

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Word-formation in Hebrew, a language of Semitic origin, is largely root-based (Berman, 1978; 2003; Ravid, 2003; Shimron, 2003). This talk considers the distribution and ratings for Familiarity and Frequency of derived nouns in Hebrew, in order to shed light on the impact of language typology on the mental lexicon. Analysis is based on 2,400 verb-related derived nouns of different historical origins and varying degrees of abstractness, most of which are constructed non-linearly from consonantal roots and affixal patterns. All the nouns were rated for Familiarity and for Subjective Frequency (Balota, Piloti & Cortese, 2001; Gernsbacher, 1984) by means of questionnaires administered to approximately 600 native speakers of Hebrew.

Closer examination reveals, first, that nearly 20% of the target nouns, in addition to having a root-based origin, are formed linearly out of other derived nouns (Ravid, 2006). For example, the noun *taklit* 'record' -- from the root *q-l-t* 'absorb, take in' combined with the affixal pattern *taCCiC* -- is the source of the recent linearly derived nouns *takliton* 'diskette' and *taklitor* '(CD ROM) disk'. Moreover, ratings for Familiarity and Frequency were significantly higher for root-based derived nouns than for linearly derived nouns. Diachronically, about half the target nouns originate from older, more classical (Biblical or Mishnaic) Hebrew and the other half from Modern Hebrew (since the late 19th century). Familiarity-Frequency ratings were also affected by this difference: Older nouns rated significantly higher scores than modern. It turns out that root-based derivation is more typical of Ancient Hebrew, while stem-external linear derivation is more typical of Modern Hebrew. Taken together, these findings reveal an interesting interaction between word-formation and diachronic processes, on the one hand, and that these intervening variables have an impact on the mental lexicon of Hebrew-speakers.

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Defective Greek nouns as evidence for multidimensional inheritance hierarchies

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Default inheritance hierarchies can capture properties common to more than one inflection class, but are not by themselves sufficient to account for all such patterns, and must be supplemented with some mechanism allowing for cross-cutting generalizations (e.g., rules of referral; Baerman et al. 2005). This raises the following question: Does default inheritance ever fail to capture the majority of inflectional patterns shared across inflection classes? In this paper I demonstrate that it is not possible to account for most of the inflectional structure of Greek nouns with a single hierarchy. However, instead of using rules of referral, I argue that lexemes participate in multiple hierarchies (alternatively, a multidimensional hierarchy), which together serve to define inflectional structure.

The core empirical problem is that, given three groups of formatives for Greek nominal inflection – singular segmental, plural segmental, and stress – any one group is often poorly predictive of the other two. I implement the full nominal system in DATR (Evans and Gazdar 1996), confirming that the mismatches among these three dimensions cross-cut each other. This entails that any one inheritance hierarchy can adequately capture the relationships among classes along only one dimension, mostly losing generalizations along the other two. I propose that lexemes simultaneously participate in all three contradictory hierarchies.

To motivate the multidimensional hierarchy, I show that a hierarchy defined according to stress patterns accounts directly for, and is independently motivated by, the distribution of genitive plural gaps within the lexicon (*kóta* ‘chicken-NOM.SG’, **kotón* ‘chicken-GEN.PL’). Yet stress cannot be the only dimension along which inheritance operates. Ultimately, I treat Greek inflectional structure as an extreme example of heteroclisis, but the extensive nature of the mismatches in Greek suggests the need for a more radical solution than paradigm linkage (Stump 2006).

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Innovative Words and Language Death: Lexical Derivation in Calabrian Greco

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One of the more interesting linguistic phenomena discussed in literature on the *dead language* condition is that of the emergence of *innovative words*, “words which use the word-formation patterns of the language, are interpretable by the context but do not appear in the standard or dialect dictionaries” (Gal 1989: 326)¹. These lexical forms are innovative not only because they are not attested in dialect dictionaries, but also because they are due to a new use of derivational devices by the speaker. I have found *innovative words* in the performance of several informants of *Greco*, a Calabrian variety of Greek, a language that has been described as ‘moribund’ since the 1950’s; the traditional enclave of the language is constituted by Amendolea, Bova, Galliciano, Roccaforte del Greco and its Chorio, Rochudi and its Chorio, and a few Aspromonte communities near Reggio in Amendolea Valley. There, the language has shifted to romance varieties (Calabrian dialect in particular) in every communicative domain; *Greco* is now an individual linguistic ‘ability’ performed by those who know it and can use it when requested; I will refer to those subjects as *semispeakers*² in a broad sense. I have collected linguistic data in the enclave through a questionnaire that ensures an exact diachronic comparison with the results of the 1924 inquiry reported in: Jaberg, K., Jud, J. (eds.) (1928 – 1940) *Sprach- und Sachatlas Italiens und der Südschweiz*, vol. 8. Zofingen. My presentation will mainly deal with the *innovative words* produced by three male informers (45-75 years old) from Bova, Galliciano and Rochudi during my interviews with them. I will explain the word formation devices involved, in particular I will focus on derivation by suffix, affix and composition in order to determine whether these devices are different from those attributed by grammars³ to the dialect when it was spoken and whether the interference mechanism played a dominant role in the innovation.

¹ Gal, S. (1989) Lexical innovation and loss: the use and value of restricted Hungarian. In Dorian, N. C. (1989) (ed.) *Investigating obsolescence. Studies in language contraction and death*. Cambridge University Press, Cambridge.

² For this problematic definition see Campbell, L. Language death. In Asher, R. E.; Simpson, J. M. Y (eds.) (1990) *The encyclopedia of Language and Linguistics*, vol. IV, 1988 - 1991. Pergamon Press, Oxford, New York, Seoul, Tokyo.

³ Mainly Rohlfs, G. (2001) *Grammatica storica dei Dialetti Italogreci* (Calabria, Salento). Mario Congedo Editore, Galatina; Καραναστάση, Α. (1997) *Γραμματική των ελληνικών ιδιωμάτων της κάτω ιταλίας*. Ακαδημία Αθηνών, Αθήναι.

Conversion: a cross-linguistic research

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In contrast to inflectional morphology, little has been done in the field of word-formation universals and word-formation typology. In fact, this field remains a more or less untilled area, and to the best of my knowledge, there has been no extensive cross-linguistic conversion-focused research either. This paper presents the data on the occurrence of conversion, and its productive use in coining new words in a sample of 65 languages of 42 genera and 23 families (classified in accordance with Haspelmath, Dryer, Gil & Comrie 2005).

The data were collected by means of a questionnaire filled out by native speakers and/or specialists in the morphology of the languages in question, and where necessary completed with the available data from grammars on these languages. The data show that conversion (in its broad sense – word- and stem-based) is not a homogeneous phenomenon. The paper points out considerable diversity of this word-formation process, including ‘homonymous’ and ‘non-homonymous’ conversion; conversion associated with stress alternation; conversion accompanied with consonant alternation and/or vowel alternation; conversion accompanied with prefixation and/or reflexive derivation, etc. The sample data indicate that an inflectional paradigm may also play an important role in the derivational process broadly labeled as conversion (cases of ‘transflexion’), thus contributing to the vague borderline between derivational and inflectional morphology. Individual ‘subtypes’ of conversion are illustrated with numerous examples.

Furthermore, the paper examines possible correlations between the occurrence/productive use of conversion and other word-formation processes; between conversion and the morphological type, the genetic type; and the geographical area of the languages covered by our sample.

In addition, the paper discusses the issue of word-classes in relation to the existence of languages where the traditional Indo-European classification does not work.

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Some remarks on clipping of deverbal nouns in French and Italian

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It is widely assumed that morphological phenomena such as clipping can be well explained within a sociolinguistic framework (as a kind of sociolinguistic variation). Nevertheless, there have been some attempts to show that there might be also specific semantic constraints that rule out the possibility of such morphological reduction. I intend to show that, as far as the restricted class of deverbal nominals in French (e.g. *introduction* → *intro*) and especially in Italian (e.g. *modificazione* → *modifica*) is concerned, special and subtle semantic restrictions seem to constrain such possibility, though the two languages do not cover exactly the same group of formations.

In French, the noun, when clipped, receives a special result/object reading although some aspects of event interpretation are maintained. The internal arguments of the base verb are, in such a formation, excluded (cf. Kerleroux, 1997: 171): a) La manifestation de la vérité aura pris cinquante ans/b) *La manif de la vérité aura pris cinquante ans/c) La manifestation (des étudiants) a duré cinq heures/d) La manif (des étudiants) a duré cinq heures.

In Italian, (cf. Thornton, 1990; 2004: 519), the shortened forms are to be taken just as stylistic variants of their corresponding full nominals. But, as pointed out by Štichauer (2006), there seem to be at least three groups of such clipped nominals that behave differently with respect to the original deverbal nouns with the suffix *-zione*. As we can see in the examples below (2) – (3), the clipping seems to be sensitive to the precise result/object meaning which is very close to what Pustejovsky (1995: 174; cf. Melloni, 2007: 27, 175) calls *information object*. It thus appears that whenever there is such an *information object*, the clipping rule may apply: (2) a) La riunificazione delle due Germanie durò molto a lungo/b) *La riunifica delle due Germanie durò molto a lungo/c) *La riunificazione è falsa/d) *La riunifica è falsa. (3) a) L'autentificazione di firme da chi non è autorizzato a farlo avviene spesso/b) ?L'autentifica di firme da chi non è autorizzato a farlo avviene spesso/c) L'autentificazione della firma è falsa/d) °L'autentifica della firma è falsa.

On the basis of more synchronic (as well as diachronic) data to be presented and analyzed, I wish to argue that the clipping rule is sensitive to the information-object meaning of the construction in *-zione*. But there seem to be two ways in which such meaning is carried by the *-zione* nominal. First, the information-object meaning can be predicted from the general semantics of the base verb. Second, the *information object* meaning aspect may not be predictable from the semantics of the base verb.

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Pleonastic morphology dies hard

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Univerbation of syntactically adjacent elements is a very common diachronic process which transforms morphologically simple words into morphologically complex ones as what formerly consisted of one morph/morpheme becomes a combination of at least two morphs/morphemes. Developments of this kind belong to the paradigm cases of grammaticalisation when erstwhile free morphemes are shown to lose their autonomy and fuse with their neighbours to turn into bound morphology of their new host. There are many pertinent examples of morphologisation in the current literature on grammaticalisation processes. Haspelmath (1993) discusses at length the problems which arise if one of the coalescing syntactic words happens to be morphologically complex already because it is inflected for some grammatical category. The evidence adduced by Haspelmath speaks in favour of what he terms externalisation i.e. if the inflectional morphology is trapped (Harris/Faarlund 2006) inside the new word form resulting from univerbation then, in the course of time, this word-internal inflection tends to be relocated and ultimately winds up (preferably) on the right margin of the word form and thus receives the status of a “normal” suffix. In my talk, I address the issue of the processes which

- (a) lead to the univerbation of syntactic words both of which are inflected for the same categories (“agreement” leading to “word-internal agreement” [Stolz 2007]) and
- (b) determine what can become of pleonastic morphology resulting from univerbation.

It is shown that externalisation is only one of the many possibilities for reorganisation of pleonastically inflected word forms. There are many possible (not necessarily only) intermediate stages which allow pleonastic morphology to be refunctionalised one way or another. I also argue that morphological pleonasm is not automatically doomed to disappear from a language’s morphological system but may survive in much the same shape for a relatively long period of time.

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Morphological or Morphosyntactic?: Variation in Intransitive Constructions in Japanese

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There are two constructions in Japanese where the choice of transitivity for the verb plays an important role. They take the forms of (a) [theme/patient]-Nom ... intransitive verb (gerund) + *iru* (e.g., *doa-ga aite-iru* “The door is open”) and (b) [theme/patient]-Nom ... transitive verb (gerund) + *aru* (e.g., *doa-ga akete-aru* “The door has been opened”). The verbs *iru* and *aru* are independently stative verbs, and the choice between the two depends on the transitivity of the preceding verb in the gerundive form. They both describe the state of the theme/patient NP as a result of the event denoted by the verb. However, we find a large number of examples in our on-line data that suggest that an innovative pairing is increasingly acceptable. The innovative forms are (c) [theme/patient]-Nom ... intransitive verb (gerund) + *aru* and (d) [theme/patient]-Nom ... transitive verb (gerund) + *iru*, where the normative pairing of an intransitive verb with *iru* and that of a transitive verb with *aru* are crisscrossed. The pattern of (c) is exemplified by *Hai-ga shizunde-aru* “Ashes are sunk” and the pattern of (d) by *Takezutsu-ga shizunde-iru* “A bamboo tube has been sunk”. The verbs in the innovative patterns in (c-d) tend to be morphologically related transitive-intransitive verb pairs

The emergence of the innovative patterns may reflect at least two possibilities: (i) the two conventional patterns in (a-b) are losing their morphosyntactic distinction, thereby allowing for the patterns in (c-d) as full-fledged variants of the normative pairing; or (ii) morphologically related transitive-intransitive verb pairs are losing their morphological distinction. To examine these possibilities, a questionnaire-based survey was given to two age-groups (20’s and over 50) for their judgments. The results show that both (i) and (ii) are taking place although the two age groups react to the phenomena differently: (i) is more significantly prevalent among the younger age group particularly in the pattern of (c); by contrast, (ii) is observed more significantly among the older group than younger speakers. Furthermore, for all speakers, both types of change, morphosyntactic and morphological, appear to be specific to a set of verbs, not applying to just any.

From class formation to event nominalization: The morphosemantic change of the French – *age*-derivation

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In this talk, I focus on the semantic development of the French *-age*-suffixation, systematizing the descriptive approach of Fleischman (1990) with the help of Lieber's (2004) division of the morphological component into 'skeleton' and 'body'. The main hypothesis is that the shift of the 'skeletal' representation of *-age* from class formation to event nominalization was induced by the suffix's ability to derive class-referring event denotations that approached it to the Old French (OFr) standard event nominalization in *-ment*.

One of the arguments in favour of the class formation analysis of OFr *-age* is its diachronic development: *-age* developed out of the Latin *-aticu*-adjectives, as in e.g. *censu agraticu* ('tribute on goods belonging to agriculture'), where the adjective serves to restrict the class of individuals denoted by the head noun (*censu*) to the subclass of individuals denoted by the derivational base. Another argument for the class-formation analysis is the systematic metonymic shift of OFr *-age*-derivatives to fiscal and juridical terms (e.g. *passage* 'tax on / right to cross a territory') and to expressions denoting a person's profession or status (e.g. *barnage* 'qualities associated with a noble'), as all these refer to the entirety of individuals denoted by the derivational base. Since OFr *-age* constrained the base to refer to classes/entireties, its semantic impact may be understood as altering the internal quantitative constitution of the base, such that the [+CI]-feature of Lieber (2004) is an appropriate representation at the skeletal layer.

As for the suffix's change from class formation to event nominalization, I argue that due to its class formation capacity, *-age* could, in principle, attach to any kind of eventive base to derive non-specific event denotations:

- (1) „*tex danrees sont en vandaige*“ (Le tournoi de Chauvency de Jacques Bretel, v. 1-2765)
'such goods are offered for sale'

By doing so, *-age* crucially approached the OFr *-ment*-derivation, which quite frequently shows up in non-specific contexts, too:

- (2) „*a nape firent parlement comment prendroient vengeance*“ (Chronique Métrique, l. 535)
'in Naples they debated how they could take revenge'

That is, while OFr *-age* and *-ment* were totally distinct as to their underlying function, (non-shifted) *-age*-derivatives frequently resembled the eventive *-ment*-nominals at the level of word usage (roughly Lieber's 'body'-level). I will argue that this overlap at the usage level facilitated the use of eventive *age*-derivatives with specific reference, which in turn caused the skeletal representation of *age* to change from class formation to event nominalization.

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Overgeneralization of irregular plural forms in the acquisition of German as a Second Language by children from Russia and Turkey.

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The paper will represent the results of a reanalysis of my data on the acquisition of German inflectional morphology by children in primary school (6 – 10 years of age) coming from Russia and Turkey (Augsburger Korpus, collected 1989-1993). The revision of the data reveals the following phenomena: not surprisingly, the most frequent plural marker of German, i.e. the suffix *-(e)n*, exhibits the highest number of overgeneralizations, however, the two plural markers occurring in combination with Umlaut (stem vowel fronting) also lead to a minor but not negligible number of overgeneralizations, such as **Söhner*, **Punkte*, **Zitröne*, which do not only occur with phonetically similar nouns, so they cannot be explained by analogy. In some cases, even regular forms are irregularized by an Umlaut (ex. *die *Häsen* /*die Hasen*) despite the fact that *-(e)n* is never combined with Umlaut in the input. However, by the criteria of frequency and productivity, the plural suffixes which occur in combination with vowel fronting such as in *Füchse* and *Bücher*, have to be considered irregular (see Wegener 2002).

These findings are interesting for the ongoing debate whether children follow a dual mechanism model in the acquisition of inflectional morphology (Clahsen 1999), which would mean that they parse and, as a consequence, overgeneralize only the regular plural marker(s), while the others are stored as whole units in lexical memory. Rather, the findings support the assumption of connectionist network models (Bybee 1995), which assume a single associative memory system that computes the different forms of a paradigm AND the different forms of a plural class and connects them by building output oriented plural schemas.

As has been shown in recent investigations of L1-acquisition data for the German noun plural (e.g. Bittner/Köpcke 2001, Szagun 2001), these GSL learners also acquire the different plural markers by using distributional and frequency information contained in the input.

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